



2024 K-Innovation Partnership Program with Azerbaijan

Policy Consultation on the Operational Programs of HTP in Azerbaijan: Implementation and Capacity Building (3rd Year): Implementation of the Application Process with the Implementation of Capacity Building of the TC Start-up and Incubation System in Azerbaijan

2024 K-Innovation Partnership Program with Azerbaijan

Title:	Policy Consultation on the Operational Programs of HTP in Azerbaijan: Implementation and Capacity Building (3rd Year) Implementation of the Application Process with the Implementation of Capacity Building of the TC Start-up and Incubation System in Azerbaijan
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Policy Consultation on the Operational Programs of HTP in Azerbaijan: Implementation and Capacity Building (3rd Year) Implementation of the Application Process with the Implementation of Capacity Building of the TC Start-up and Incubation System in Azerbaijan

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Executive Summary

The “Implementation of the Application Process and Capacity Building of the TC Start-up and Incubation System in Azerbaijan” project represents a strategic initiative aimed at strengthening Azerbaijan’s innovation ecosystem. This effort seeks to establish a robust start-up and incubation framework supported by comprehensive capacity-building programs, thereby addressing critical gaps and positioning Azerbaijan as a regional hub for innovation.

In recent years, Azerbaijan has made significant progress in diversifying its economy beyond its traditional reliance on sectors such as oil and gas. The Technology and Creativity (TC) sector has emerged as a key pillar of this diversification strategy, reflecting the country’s aspiration to foster a dynamic and innovative economic environment. Despite its potential, the TC sector faces challenges, including inadequate infrastructure, limited expertise, and misaligned policy frameworks, which have hindered the growth of start-ups and incubation systems. In response, this project was conceptualized to address these barriers and cultivate a thriving entrepreneurial ecosystem in Azerbaijan.

The primary objectives of the project include enhancing the skills and expertise of key stakeholders through targeted capacity-building initiatives and aligning institutional policies to support sustainable innovation. These goals have been operationalized through a series of carefully planned activities aimed at driving meaningful change across the ecosystem.

The project began by engaging stakeholders to understand the specific needs of Azerbaijani start-ups. This engagement informed the design of a digital application platform, developed to ensure transparency and accessibility. The platform was piloted with a select group of start-ups to gather feedback and subsequently refined to improve its functionality. Complementing this initiative, tailored capacity-building programs were introduced to

empower incubator staff, mentors, and entrepreneurs. Training sessions covered key topics such as best practices in mentorship, resource management, business model development, fundraising, and scaling strategies. Korean experts were invited to share their insights, fostering a culture of innovation and global connectivity.

To address policy-related challenges, the project undertook a comprehensive review of existing frameworks, identifying bottlenecks and proposing actionable reforms. Collaboration with government agencies led to streamlined licensing and regulatory processes, reducing administrative burdens on start-ups. Mechanisms for monitoring and evaluating the effectiveness of incubation initiatives were also established to ensure continuous improvement. International collaboration was prioritized, enabling Azerbaijani start-ups to access mentorship opportunities, participate in global expos and competitions, and benefit from exchange programs that facilitated knowledge sharing with international stakeholders.

The project has achieved significant milestones, underscoring its transformative impact. More than 200 stakeholders, including incubator managers and entrepreneurs, have completed capacity-building programs, equipping them with the skills required to thrive in a competitive market. Policy reforms have streamlined bureaucratic procedures, reducing barriers for start-ups and fostering a more conducive environment for innovation.

While the project has been largely successful, it has encountered challenges that offer valuable lessons for future initiatives. Resistance to change from traditional institutions required targeted engagement strategies, while limited digital literacy among some stakeholders necessitated additional training. Bureaucratic hurdles delayed the implementation of some policies, emphasizing the need for persistent advocacy and adaptive approaches. These experiences highlight the importance of early stakeholder involvement, iterative design processes, and robust monitoring mechanisms to ensure the sustainability and scalability of interventions.

Overall, the “Implementation of the Application Process and Capacity Building of the TC Start-up and Incubation System in Azerbaijan” project has laid a solid foundation for the

country's TC ecosystem. By addressing critical gaps, empowering stakeholders, and fostering collaboration, it has positioned Azerbaijan to harness the full potential of its TC sector and emerge as a leader in regional innovation. The lessons learned from this initiative will serve as a valuable guide for future efforts aimed at promoting sustainable economic growth through innovation and entrepreneurship.

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CHAPTER 1

Project Overview

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Chapter 1. Project Overview

1. Project Background

According to the definition provided by the World Intellectual Property Organization (WIPO), technology transfer plays a pivotal role in facilitating the entire life cycle of a technology, encompassing its inception, diffusion into the market, and eventual commercialization¹. Within this context, the commercialization of technologies entails a process in which an idea transitions from a knowledge base to the market, thereby creating financial value. Technology commercialization (TC) also enhances economic efficiency by offering consumers a diverse and improved range of goods and services, resulting in the generation of value-added revenue streams. Consequently, investment in technology and research and development (R&D)-related activities, along with the subsequent commercialization of their outcomes, necessitates the involvement of a wide array of institutions, programs, and mechanisms to achieve the optimal results.

In the context of Azerbaijan, technology commercialization (TC) has recently emerged as a prominent topic of discussion within its innovation ecosystem. This report is an integral part of a collaborative project between the Azerbaijan High Technologies Park (HTP) and the Science and Technology Policy Institute (STEPI) spanning 2022 to 2024. The report's primary focus is to delve into the nuances of technology transfer and commercialization that are specific to Azerbaijan. To accomplish this objective, our research will explicitly centre on the developments occurring in Azerbaijan post-July 2022, coinciding with significant institutional reforms in the aforementioned domains. We anticipate that this report will contribute significantly to enhancing the effectiveness of the STEPI-HTP partnership's operational programs.

¹ <https://www.wipo.int/technology-transfer/en/index.html>

Consistent with the WIPO's delineation, effective technology transfer requires universities and research institutions to operate within a well-structured innovation ecosystem². This ecosystem comprises an interconnected network involving governmental bodies, industry stakeholders, research institutions, and various enabling factors, including human capital, technology transfer infrastructure, and the sophistication of businesses and markets. Within such an ecosystem, these entities pool their resources and expertise to collaboratively drive innovation, ultimately serving regional and economic development objectives.

Recognizing these imperatives, Azerbaijan has established a comprehensive framework of institutions and infrastructures to regulate and bolster TC activities. For instance, the "Social-Economic Development Strategy of the Republic of Azerbaijan, 2022-2026" outlines a strategic framework, part of which is dedicated to enhancing the competitiveness of the country's human capital and positioning it as a hub for innovation. Consequently, Azerbaijan aspires to create a "Science-Industry Cluster 4.0" concept-SABAH.city. SABAH.city is a transformative innovation cluster slated for launch in 2026, driven by the Ministry of Science and Education in collaboration with key public and private stakeholders. The initiative aims to create a thriving ecosystem that fosters research, development, and innovation, positioning SABAH.city as a major regional hub. The development is planned across three strategic phases:

Phase 1: 2024-2026 - Laying the foundations and building initial traction

During this foundational phase, SABAH.city will begin establishing initial facilities across approximately 4 hectares, setting the stage for future growth. Key components, including startup incubators, accelerators, venture studios, venture capital funds, and technology transfer offices, will be operational, providing essential support for emerging businesses. In addition, leading public and private educational institutions and pioneering companies will relocate and establish themselves as the initial residents of SABAH.city, contributing to a vibrant community focused on innovation.

² <https://www.wipo.int/technology-transfer/en/index.html>

Phase 2: 2026-2029 - Scaling up and expanding capabilities

In the second phase, the developed area will expand to roughly 15 hectares, enabling the construction of additional infrastructure and enhanced facilities. This phase will introduce cutting-edge research centers, state-of-the-art data centers, and advanced co-working spaces designed to foster collaboration and drive innovation. By this stage, SABAHA.city will further integrate leading educational institutions and innovative companies as foundational residents, cementing its role as a dynamic environment for research and technological advancement.

Phase 3: 2029 and Beyond - further expansion and internationalization

In the final phase, SABAHA.city will extend across the full 20 hectares, maximizing resource utilization and spatial efficiency. This expansion will include welcoming leading international companies seeking to establish regional offices, thereby expanding the city's global reach. With this growth, SABAHA.city will solidify its position as a premier hub for research, development, and innovation, attracting talent and investment from across the globe.

SABAHA.city promises to be a landmark destination for innovation, bringing together academia, industry, and investors in a collaborative environment that supports sustainable growth and technological breakthroughs.

SABAHA.city provides a comprehensive ecosystem designed to support innovation and growth for startups and technology companies. Their offerings include:

- Access to talent: Connecting businesses with skilled professionals across various fields.
- Media coverage and PR support: Helping startups gain visibility and promote their stories.
- Networking with tech companies and startups: Fostering connections within the tech community for partnerships and growth opportunities.
- Startup and tech academies: Educational programs to develop essential skills for the startup and tech industries.
- Technology transfer programs: Facilitating the transfer of cutting-edge technology from research to market.

- Venture studios: Providing resources and expertise to develop new businesses from the ground up.
- Incubation and acceleration programs: Offering structured support for early-stage startups to scale and succeed.
- Offices and co-working spaces: Flexible workspace options designed to foster collaboration.
- Sharable research labs: Access to high-quality labs equipped for research and development.
- Tax incentives: Financial benefits to support startup growth.
- Pre-seed & seed financing and venture funds: Funding opportunities for startups at various stages.
- Corporate financing options: Financial support for established tech businesses looking to expand.

SABAH.city brings together resources, expertise, and a collaborative environment to accelerate innovation and support the entire lifecycle of tech startups and companies.

The K-Innovation Partnership Program, a collaboration between the Science & Technology Policy Institute (STEPI) of the Republic of Korea and the High Technologies Park (HTP) of the Republic of Azerbaijan, spanning 2022 to 2024, is primarily geared towards capacity building and fostering cooperation in the realms of effective technology transfer and commercialization. Within the ambit of this project, this document has been tasked with delivering a status report concerning TC policies, programs, and projects in Azerbaijan. To fulfill this mandate, the report will serve as a comprehensive reference, providing in-depth insights into governance mechanisms, institutional and legal frameworks, intellectual property rights, and a concise overview of Azerbaijan's national innovation ecosystem. Its overarching objective is to make a substantial contribution to the K-Innovation Partnership Program by offering a comparative analysis and timely updates on the pertinent reforms and policies in Azerbaijan.

2. Project Objectives

The primary objective of the 2024 initiative is to offer policy advisory services concerning the enhancement of the start-up and incubation ecosystem in Azerbaijan, along with the establishment of a start-up and incubation program tailored for the High Technologies Park (HTP). Additionally, the initiative aims to foster mutually advantageous economic collaboration between the two nations, thereby aiding the advancement of Science, Technology, and Innovation (STI) in Azerbaijan in accordance with the principles outlined in the 『FRAMEWORK ACT ON INTERNATIONAL DEVELOPMENT COOPERATION. 』 in the Republic of South Korea.

[Table 1-1] Framework Act on International Development Cooperation

Article 3. Basic Ideas and Objectives	
(1) The basic objectives of international development cooperation are to reduce poverty in developing nations, improve the human rights of women, children and people with disabilities, achieve gender equality, realize sustainable development and humanitarianism, promote mutually beneficial economic relationships with cooperation partners, and pursue peace and prosperity in the international community.	
(2) The objectives of international development cooperation shall lie in achieving the following matters so as to implement the basic ideas presented under the foregoing paragraph (1):	
The Basic Ideas of International Development Cooperation	Development of developing nations and improvement of the systems and conditions for such development.
	Promotion of friendly relationships and mutual exchanges with developing nations.
	Contribution to the resolution of global problems related to international development cooperation.

Source: Framework Act on International Development Cooperation, No.17302(2020), Republic of Korea.

- Project Objectives: 1st year (2022) - 3rd year (2024)
 - (2022) Improvement of the TC System in Azerbaijan and Development of the TC Programme at ANAS-HTP
 - (2023) Improvement of the Start-up and Incubation System in Azerbaijan and Development of the Start-up and Incubation Programme for HTP
 - (2024) Implementation of the Application Process with the Implementation of Capacity Building of the TC Start-up and Incubation System in Azerbaijan

3. Project Framework

The 2024 project is focused on a developing start-up and incubation programmes for the High Technologies Park (HTP) and improving the technology commercialization system in Azerbaijan.

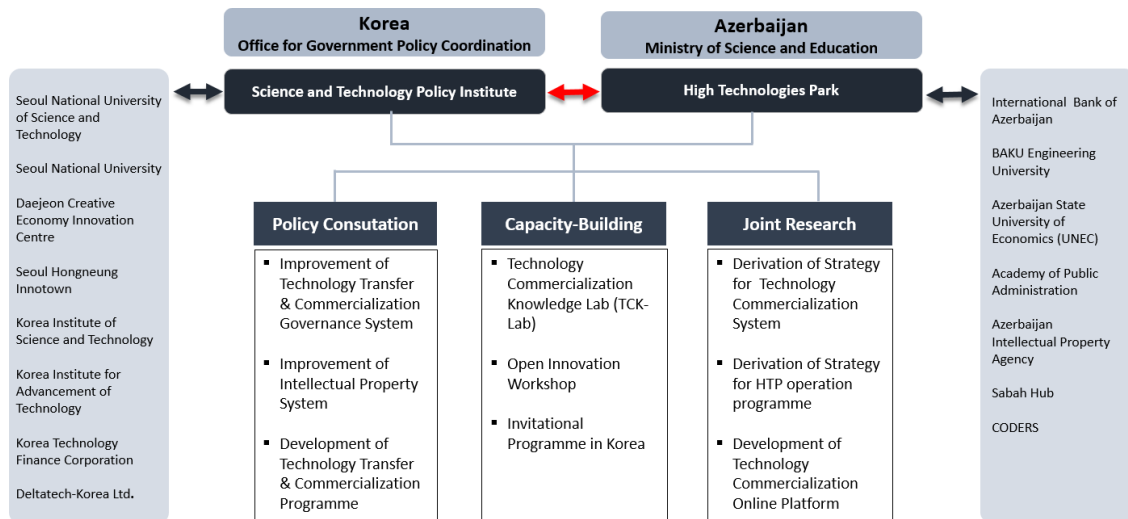
This project began with a kick-off seminar and an extensive current status analysis. During this phase, we formed a dedicated research team and outlined the scope of our research. This part of the project consisted in investigating the local landscape by conducting a thorough assessment of the existing start-up and incubation regulations in Azerbaijan, while simultaneously examining the start-up and incubation services offered at HTP, pinpointing areas that required further attention.

In the pursuit of enhancing capacity, a collaborative effort with diverse experts and organisations in South Korea was initiated. This collaboration included inviting Azerbaijan experts for training and organising specialized workshops on technology transfer and commercialization. These initiatives aimed to foster exchanges of knowledge and share Korea's experience and methodologies.

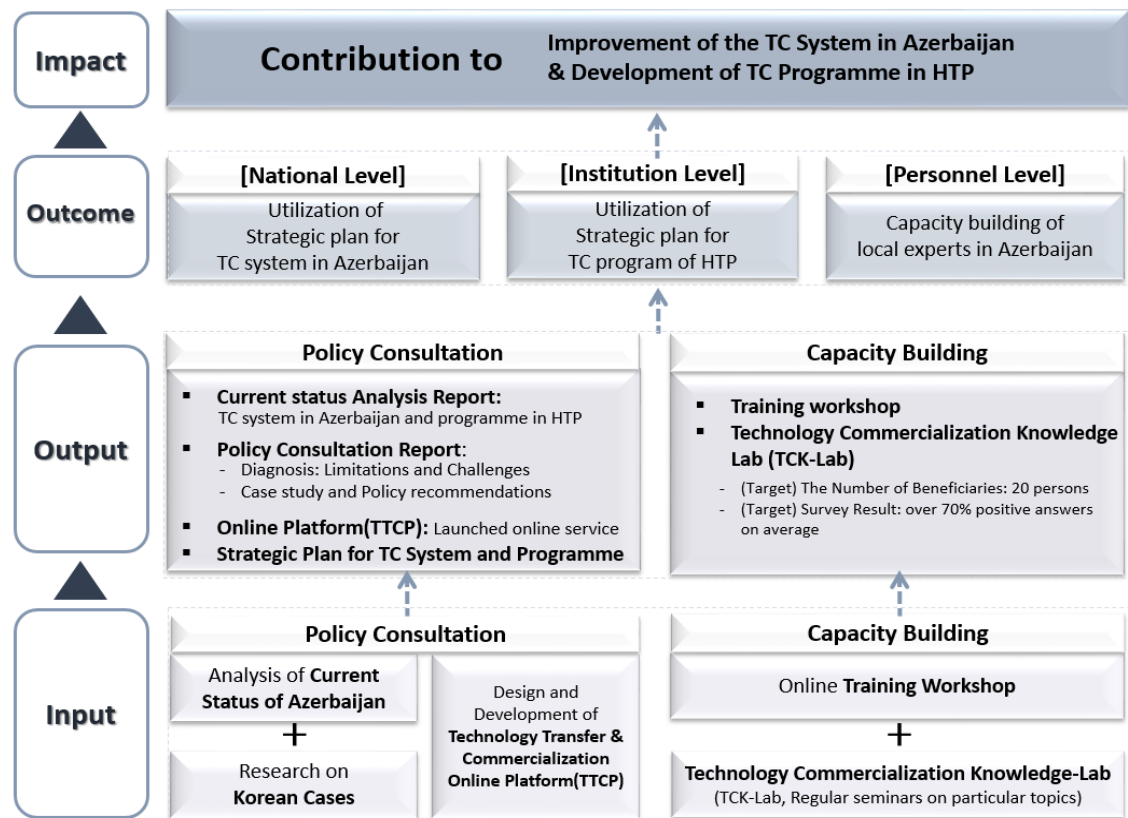
An integral aspect of the 2024 project involved the development of an online platform for technology transfer and commercialization. This platform was designed to enhance the ecosystem of technology transfer and commercialization in Azerbaijan by offering information, incorporating enhancements in content and technological functionality, and ensuring that it would serve as a valuable resource for stakeholders.

Finally, a forum has been conducted to disseminate the outcomes of the 2024 project. This forum played a pivotal role in providing policy recommendations concerning technology commercialization governance in Azerbaijan and boosting the roles of the HTP. The aim was to ensure that the insights gained from capacity-building activities were not only shared but also effectively disseminated to the relevant stakeholders, thereby contributing to the broader ecosystem of technology transfer and commercialization in Azerbaijan.

[Figure 1-1] Schematization of the 2024 Project Structure



[Figure 1-2] Detailed Framework of the 2024 Project



4. Project Team

The project team is composed of four key groups: the project management team, Korean consultants specializing in policy recommendations, Korean experts focusing on the TCK-Lab, and a team of Azerbaijan experts. The project management team oversees overall project coordination, resource allocation, and timeline management; the Korean consultants offer policy insights, aligning international standards with local needs; the Korean TCK-Lab experts facilitate knowledge transfer through seminars and workshops; and, last but not least, the Azerbaijani experts contribute local insights and conduct situational assessments to tailor the project activities. In summary, this multidisciplinary team combines international expertise with local insights to enhance Azerbaijan's technology commercialization system and foster innovation-driven growth.

[Table 1-2] Project Team

Type	Name	Position/Affiliation	Role/Area of Expertise
Project Management	Eun Joo Kim	Principal Researcher, STEPI	Project Manager
	Yoon sil Moon	Researcher, STEPI	Project Coordinator
Korean Consultants	Soo J. Sohn	Senior Researcher Fellow, STEPI	TT&TC governance
	Youngtack Shim	Professor, Seoul National Univ.	Intellectual property system
	Gyeongsoo Shim	Professor, Seoul National Univ. of S&T	TT&TC, Korean I-Corps programs
	Youngtack Shim	Professor, Seoul National Univ.	TCK-Lab (IP Silk Road)
	Seung Ho Lee	CEO, Deltatech-Korea	Open innovation workshop
	Roy Hwang	CEO, INSIGHTORS	Business Model workshop

Type	Name	Position/Affiliation	Role/Area of Expertise
Azerbaijan Experts	Bünyamin Seyidov	Head, Incubation and Technology Transfer Centre, HTP	Project Coordinator, strategic planning for programme
	Isa Gasimov	Director, Baku Engineering University and Technopark	TC platform management, strategic planning for the TC system
	Vusal Suleymanli	Innovation Consultant, ANAS	Management of TCK-Labs
	Samir Veliyev	Senior Economist, Azerbaijan International Bank	Current status research

5. Project Activities

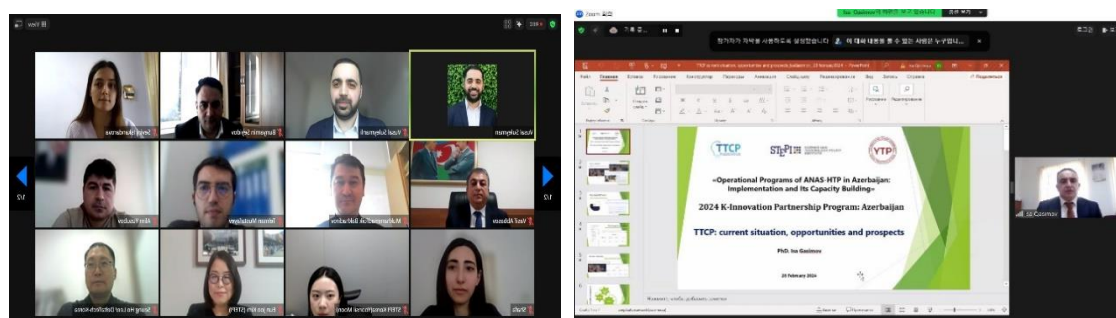
5.1 Kick-off Seminar

- Date/Venue: 15:00-17:00(Korea time), 28 February 2024 /Online
- Participants: 20 persons (STEPI and HTP Project Team, Korea-Azerbaijan experts)
- Purpose: To establish a consensus and foster mutual comprehension regarding the project's subject matter, strategic blueprints, and goals
- Contents:
 - Project Objectives and Detailed Plan, Current Status
 - Roles and Responsibilities of the Project Teams

[Table 1-3] Kick-off seminar program

Time	Contents	Presenters
15:00-15:15	Opening Remarks	Dr. Eun Joo Kim, Principal Researcher, STEPI
	Welcome Remarks	Dr. Zaur Nahmatov, Executive Director, HTP
Review and Plans for Policy Consultation		
15:10-15:30	<Review> 2023 K-Innovation Key activities	Mr. Bunyamin Seyidov, Head, Incubation and TT Centre, HTP
15:30-15:45	<Status Update> Technology Transfer & Commercialization	Samir Veliyev, Local expert
15:45-16:00	<Plan> TTCP : current situation	Dr. Isa Gasimov, Director, Technopark, Baku Engineering University
15:30-15:40	<Plan> TTT Programme	Prof. Gyeongsoo Shim, Seoul National University of S&T
15:40-15:50	Q&A	Prof. Gyeongsoo Shim, Seoul National University of S&T
15:50-16:00	Wrap-up	Dr. Eun Joo Kim, Principal Researcher, STEPI

[Figure 1-3] Kick-off seminar photos



5.2 (Capacity-Building) Open Innovation Workshop

In light of the increasing demand for innovative policy-making capabilities, a broad spectrum of diverse learning methodologies has emerged as a viable option in contemporary contexts. One such methodology that warrants consideration is the workshop approach.

The Open Innovation Workshop was conceived and executed with the overarching objective of aligning with the objectives of the 2024 project. This project's primary mission was to fortify the operational prowess and competitive standing of the HTP (High Technologies Park) through Implementation of the Application Process with the Implementation of Capacity Building of the TC Start-up and Incubation System in Azerbaijan. Additionally, it sought to facilitate the exchange and dissemination of knowledge among local stakeholders.

The Open Innovation Platform Workshop for the year 2024 represents an integral component of the capacity-building initiative within the purview of the K-Innovation Partnership Program. The workshop's ultimate aim was to furnish the participants with TC start-up basic principles and illuminate the commercial dimensions of technology, to be especially based on Korean I-Corps TTT(Train The Trainer) Program. The diverse participant base included individuals from various professional backgrounds, including current and prospective inventors, entrepreneurs, experts in technology commercialization and transfer, as well as professionals affiliated with Technology Transfer Organizations (TTOs), among others.

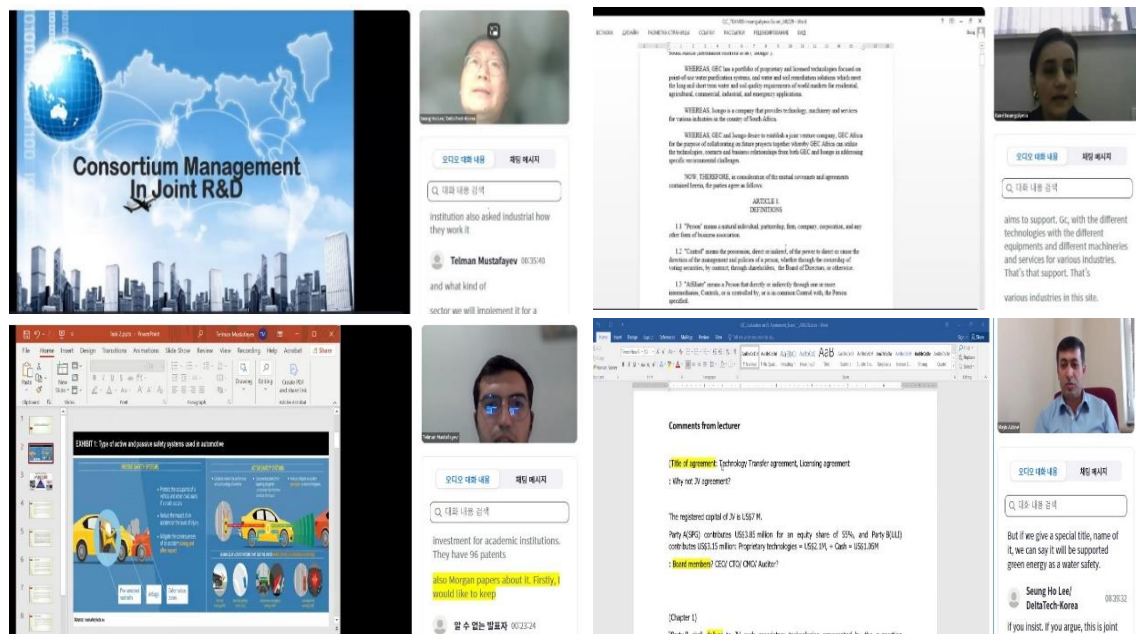
Through the strategic deployment of this workshop, it is anticipated that a multifaceted approach encompassing learning, practical application, innovative discovery, and the exchange of ideas will contribute significantly to the capacity-building efforts of ecosystem stakeholders. As a result, this initiative is poised to catalyse improvements and advancements within Azerbaijan's Technology Transfer (TT) and Technology Commercialization (TC) landscape through training the instructors.

- Date/Venue: February to March 2024 / Online
- Lecturer: Mr. Seung Ho Lee, CEO of DeltaTech Korea
- Overview: A series of four workshops aimed at enhancing open innovation
- Participants: Targeted at the Expert of the Azerbaijan Technology Commercialization

[Table 1-4] Open Innovation Workshop Curriculum

	Date	Session
1	2 February	Portfolio Mining & Due Diligence
2	15 February	Open Innovation Deal Structure
3	29 February	Consortium Management In Joint R&D
4	18 March	Community Building for Tech Transfer

[Figure 1-4] Participants in the Open Innovation Platform Workshop



5.3 (Capacity-Building) Invitational Programme in Korea

- Date/Venue: 26 – 30 August 2024 / Korea
- Purpose
 - To make study visits to major technology start-up institutions in Korea and to enhance the capabilities of Azerbaijan's technology start-up programme
 - To establish a network between Korean and Azerbaijani experts in technology start-up and industrial institutions
- Participants: 6 individuals, selected from the workshop, including experts related to technology commercialization, intellectual property, and technology start-up
- Key Contents
 - Visit prominent technology start-up institutions in South Korea to strengthen Azerbaijan's capabilities for technology transfer
 - Hold an workshop designed to Korean I-Corps "Train the Trainer" program
 - Promote networking among technology start-up experts and institutions from South Korea and Azerbaijan

[Table 1-5] Invitational Programme in Korea

Module	Type	Details
1. Technology start-up training	Lecture	Seminar on sharing experience of foreign technology Start-up
	Lecture	Seminar on Korea I-Crop and Experience in Corporate Activities
2. Train the Trainer(TTT) Program	Workshop	Kick-off Session 1 1. Registration/Surveys/Opening 2. Orientation/Introductions 3. Teaching Philosophy of Instructor, why Korean I-Corps? 4. Value Propositions Review 5. Team based Rocket Pitching/Practical session for designing VPC (Mentor Team 1: Prof. Youngtaek Shim / Mr. Seungho Lee, Team 2: Prof. John Shim / Mr. Roy Hwang)
	Workshop	Kick-off Session 2 1. Customer Discovery - Part 1 2. Customer Discovery - Part 2 (Mentor Team 1: Prof. Youngtaek Shim / Mr. Seungho Lee, Team 2: Prof. John Shim / Mr. Roy Hwang) 3. Team Presentation 4. Customer Discovery - Part 3(Full customer discovery template and interview form Introduction)
3. Korea- Azerbaijan Networking	Study Visit	HABSIDA, Education Company
	Study Visit	Embassy of the Republic of Azerbaijan

2024 K-Innovation Partnership Program with Azerbaijan

[Figure 1-5] Invitational Programme in Korea



TTT program _Mentoring



TTT program _Team Presentation



Completion Ceremony



Invitation Training Participants



HABSIDA, Education Company



Embassy of the Republic of Azerbaijan

5.4 Dissemination Forum

Dissemination workshop commenced with introductory remarks and an overview of the workshop's objectives. Dr. Eun Joo Kim, Principal Researcher, Division of Global Innovation Strategy Research (STEPI), highlighted the goals of fostering international innovation partnerships.

Mr. Nihat Majidov, Chief Director, High Technology Park (HTP), emphasized the importance of Azerbaijan's collaborative efforts in innovation and technology commercialization. Mr. Bunyamin Seyidov, Head of Department, HTP, provided insights into recent achievements and future directions of HTP's projects in advancing technology commercialization.

Session 1 focuses on open innovation, startup education, and business modeling for technology ventures. Mr. Seong Ho Lee, CEO of DELTATECH-KOREA, presented the strategies to leverage open innovation in Research & Innovation (R&I) projects, facilitating collaborations and expanding access to resources. Prof. John Shim, from Seoul National University of Science and Technology, presented on the role of education in preparing researchers and educators to guide and inspire startup culture. Mr. Roy Hwang, CEO of INSIGHTORS, stressed the critical role of structured business models in ensuring the viability and scalability of tech startups.

Session 2 dedicated to exploration of intellectual property, training programs, and the role of international ecosystems in innovation. Prof. Youngtack Shim, from Seoul National University, discussed the relative advantages of patents versus trade secrets, providing guidance on when each approach is most appropriate for protecting innovations. Ms. Gunel Imamgulyeva, from Azerbaijan University of Architecture and Construction, introduced a training initiative aimed at building local capacity in innovation coaching. Dr. Soo J. Sohn, Senior Research Fellow at STEPI, elaborated on frameworks to support sustainable innovation growth through international partnerships. Dr. Isa Gasimov, Director of Baku Engineering University Technopark, outlined how elements of Korea's innovation ecosystem can serve as a model for fostering technological and economic development in Azerbaijan.

2024 K-Innovation Partnership Program with Azerbaijan

- Date/Venue: 23 October 2024 /Baku Azerbaijan
- Participants: 50 stakeholders in Azerbaijan
- Purpose: To conduct policy advisory and capacity enhancement activities in the implementation of the K-Innovation Partnership Programme for Azerbaijan's "Project to Improve the Start-up and Incubation System in Azerbaijan and Develop the Start-up and Incubation Programme for the High Technologies Park (HTP)" and to disseminate the outcomes.

[Table 1-6] Dissemination Forum Programme

Time	Contents	Presenters
	<Opening Session>	
10:00-10:20	Opening Remarks	Dr. Eun Joo Kim Principal Researcher, Division of Global Innovation Strategy Research (STEPI)
	Welcome Remarks	Mr. Nihat Majidov Chief Director, HTP
10:20-10:30	"Operational Program of HTP: Implementation and Capacity Building" project: Outcomes and Impacts"	Mr. Bunyamin Seyidov Head of Department, HTP
	Photo session	
	Session 1	
10:30-10:45	How to Initiate R&I Project via Open Innovation (OI) Platform	Mr. Seong Ho Lee CEO / DELTATECH-KOREA
10:45-11:00	Design and Development of Technology based Startup and Education Program for Educators and Researchers in Azerbaijan	Prof. John Shim Seoul National University of S&T
11:00-11:15	Why Business Models are Important for Tech Start-ups and How We Approach	Mr. Roy Hwang CEO / INSIGHTORS
11:15-11:35	<i>Coffee break</i>	

Time	Contents	Presenters
Session 2		
11:35-12:00	Patents or Trade Secrets? Two Opposite Models for protecting your Business Models	Prof. Youngtack Shim Seoul National University
12:00-12:15	"Train the Trainers" journey (K-Innovation Partnership Program with Azerbaijan, 2024)	Ms. Gunel Imamgulyeva Azerbaijan University of Architecture and Construction
12:15-12:30	Bridge for Innovation Growth, Technology Commercialization	Dr. Soo. J. Sohn Senior Research Fellow (STEPI)
12:30-12:45	The role of Korean experience in the innovation ecosystem of Azerbaijan	Dr. Isa Gasimov Director, Baku Engineering University Technopark
12:45-13:00	Q&A /Closing Ceremony	Mr. Vusal Suleymanli

[Figure 1-6] Dissemination Forum



6. Project Schedule

[Table 1-7] Project Schedule

Activities	2024												2025
	1	2	3	4	5	6	7	8	9	10	11	12	1
Project Planning	■												
Design Project Framework	■												
Organizing Project Team	■	■											
Kick-off Seminar		■											
Open Innovation Workshop			■	■	■								
Invitational Programme								■	■				
Analysis of Current Status					■	■	■	■					
Diagnosis of Current Status						■	■	■	■				
Policy Consultation				■	■	■	■	■	■	■			
Establishment of Strategic Plans										■	■		
Dissemination Forum										■	■		
Submission of Final Report											■	■	
Presentation of Final Report												■	■
Evaluation of Final Report												■	■
Publication													■



CHAPTER 2

Current Status of Technology Commercialization in Azerbaijan

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Chapter 2. Current Status of Technology Commercialization in Azerbaijan

1. Research Framework

A systematic review of existing academic literature on technology commercialization can demonstrate the ambiguous role of the capabilities of technology transfer offices (TTO) in transferring technologies from a university/research institute to industry. For example, Lee and Jung (2021) identified three distinctive capabilities of technology commercialization (TC) while analysing longitudinal data from South Korea for the period 2009-2015. These capabilities include academic research capability, applied research capability, and TTO operation capability. Their analysis of the effects of these capabilities on TC performance reveals that, unlike the other two capabilities, the TTO operation capability has no significant effect on TC performance. Therefore, establishing an effective operational capability alone for a high-technology park cannot solve the challenges associated with TC activities. Min, Kim and Vonortas (2020) also analysed the conditions for TC and commercial exploitation of public technologies in Korea. Their analysis of 514 cases of TC by major public research institutes and universities underline that market competition is a key determinant of successful TC, and that it also affects business growth in general.

Moreover, regardless of the market situation, effective partnership is critical for successful TC. Recently, Belitzki, Aginskaya and Marozau (2019) also identified the role of TTOs and direct industrial funding in TC in the transition economies of Azerbaijan, Belarus and Kazakhstan during 2015-2017. Interestingly, their research reveals that successful research commercialization is neither associated with the existence and awareness of TTOs, nor with their contracts with partners. However, direct industrial funding of research is a significant factor.

1.1 Macro-level analysis of the current status: a view of the TC ecosystem in Azerbaijan

■ Policy-making issues

A successful innovation policy for a developing country requires careful consideration of a wide range of activities. To achieve better TC results, policy makers should develop a comprehensive innovation policy and then consider certain key issues when implementing that policy. Such issues may include, but are not limited to, the following: the state of innovation; strategic priorities; access to finance; intellectual property protection; capacity building; and collaboration and cooperation.

■ Institutions

TC is a process by which necessary measures are undertaken in order to pass a given research output through the relevant procedures and make it beneficial for commercialization in various applications. For TC to act effectively, several types of institutions play important roles: government agencies; research centres and universities; business incubators and accelerators; venture capital firms; and associations and consortia.

■ Funding mechanisms

TC requires several funding mechanisms to be available. The range of such funding schemes may vary depending on the stage of technology development, or on the goals of the stakeholders who seek to commercialize such technologies. Various major funding mechanisms also play an important role, including government grants, equity financing, debt financing, corporate partnerships, and crowdfunding.

■ Regulatory frameworks

Complying with legislation and regulations is a must-do requirement for TC. Therefore, it is critical to assess regulatory frameworks that may impact it. There are several regulatory frameworks that are particularly important for TC, and such frameworks - encompassing

intellectual property, taxation, international trade, consumer protection, data privacy, and environmental, health and safety regulations among others - may vary depending on the specific industry and technology involved.

1.2 Micro-level analysis of the current status of the HTP

■ Operational issues

Improving the operational issues related to TC, particularly in high-technology parks, involves different strategies and actions. By improving them, stakeholders can achieve a better assessment of the status quo, develop appropriate and effective strategies and plans, and monitor their implementation and results in later stages. There are certain operational issues that are critical to TC, i.e. needs assessment, improvement plans, infrastructure and services, collaboration and cooperation, and monitoring and evaluation.

■ Other issues

Along with the abovementioned elements of a rational framework of analysis for this report, the following other issues are also important to consider when aiming for successful TC: market potential, business model, timing, team, etc.

Based on the research framework described above, this report will analyse the current status of TC in Azerbaijan.

2. Macro-Level Analysis of Technology Commercialization in Azerbaijan

A successful innovation policy for a developing country requires careful consideration of a wide range of activities. In order to achieve better TC results, policy makers should develop a comprehensive innovation policy and then consider certain key issues when implementing that policy. Gaining added value through TC plays a key role in economic development. The rapid development of technology and increasing competition are among the main factors that justify the commercialization of science, as well as the development of innovative entrepreneurship. There are complex infrastructural factors in Azerbaijan to regulate the process of TC. At the same time, there are factors that restrict this process and partially hinder it, such as legislation, the weakness of university-industry cooperation, incomplete formation of the intellectual property ecosystem, and other factors. Also, the lack of research that meets the needs of the industry can be considered as a factor that slows down the commercialization process.

The Global Innovation Index (GII) for 2023 was compiled based on 81 indicators that cover the potential, efficiency, and framework conditions of innovation activities. The analyzed indicators encompass a wide range of information, from legislation, science, and education to the development of markets, business, infrastructure, and more. The main purpose of publishing the GII is to evaluate the efficiency of countries' innovation activities.

Switzerland leads the 2023 GII ranking, followed by Sweden, the United States, the United Kingdom, and Singapore. The top ten most innovative countries also include the Netherlands, South Korea, Germany, Finland, and Denmark. China ranks 12th, entering the second group of countries.

Among post-Soviet countries, Estonia has the best results, ranking 16th, followed by Lithuania at 34th and Latvia at 47th. Russia is ranked 51st.

In the 2023 ranking, Azerbaijan improved its position by four spots compared to 2022, ranking 89th. It should also be noted that while last year Azerbaijan was in the group of countries with comparative indicators starting from 90th place, in 2023, it is in the group of countries starting from 85th place.

The GII's expert compilers highlight the role of government bodies as one of Azerbaijan's strengths, reflected in the "Institutions" pillar, where Azerbaijan ranks 42nd. Azerbaijan ranks 87th in the "Human Capital and Research" pillar. This year, Azerbaijan ranks 76th in the education index, which is 16 positions better than last year. Within this pillar, the country ranks high in the "student-teacher ratio" at 17th, "government expenditure on students" at 28th, "researchers" at 44th, and "graduates in science and technology" at 47th. The composite indicator for "Research and Development" improved from 76th place in 2022 to 73rd place in 2023.

Azerbaijan ranked 95th in the GII 2024, reflecting both the progress and areas for improvement in the country's innovation ecosystem.

The GII is based on two main sub-indices: "innovation resources" (inputs) and "innovation results" (outputs). Azerbaijan performed well in certain areas, particularly in government efficiency, business development, and industry-university collaboration.

Key highlights include:

- **Institutions:** Azerbaijan ranked 51st in government efficiency, regulation quality, and political stability.
- **Business Environment:** The country rose to 15th place, reflecting an increasingly favorable climate for innovation and business development. Azerbaijan also advanced in business policy stability, securing 19th place.
- **Business Development:** Ranked 67th, Azerbaijan's performance in this area highlights ongoing improvements in economic infrastructure and market conditions.
- **Scientific Research and Cluster Development:** Azerbaijan leads the South Caucasus in research and cluster development, ranking 30th and 29th respectively, indicating strong

collaboration between academia and the private sector. This is aligned with Azerbaijan's 2030 development agenda, emphasizing industrial clusters and R&D cooperation.

Azerbaijan made notable strides in patent activity, rising to 50th place in "Knowledge and Technological Results" and leading the South Caucasus in this category. The country also improved in labor productivity, climbing 34 spots to 28th place, thanks to the adoption of advanced technologies and improved workforce skills.

In terms of human capital and science, Azerbaijan ranked 94th, but significant progress was observed in areas like student-teacher ratio (20th) and government investment in education (53rd). The country's 45th-place ranking for science and engineering graduates highlights its growing technical talent base, which is essential for innovation in key sectors like technology and engineering.

However, experts emphasize the need for more venture capital, increased high-tech exports, and greater access to private sector credit. These steps, along with stronger collaboration between education and industry, will be crucial for further boosting Azerbaijan's innovation capacity.

In the first half of this year, KOBIA (Small and Medium Business Development Agency) issued "Startup Certificates" to 38 startups. These startups primarily cover fields such as finance, education, medicine, tourism technologies, e-commerce, robotics, and other areas.

As part of the goals set for 2024 under the "Strategy for Socio-Economic Development of the Republic of Azerbaijan for 2022-2026," the activities of the "Innovation" Working Group have been continued in an improved manner. The "Innovation" Working Group includes representatives from the Ministry of Digital Development and Transport, the Ministry of Economy, the Ministry of Science and Education, the Ministry of Foreign Affairs, the Ministry of Finance, the Ministry of Labor and Social Protection, the Ministry of Justice, the Ministry of Defense Industry, the Ministry of Agriculture, the Ministry of Culture, and the Ministry of Energy. Additionally, it includes the Central Bank, the State Customs Committee, the State Committee on Work with Diaspora, the State Agency for Public Service and Social

Innovations under the President of the Republic of Azerbaijan, the State Water Resources Agency, the State Migration Service, the State Border Service, the Intellectual Property Agency, and relevant departments and sections of the National Academy of Sciences (ANAS).

It should be noted that with the Decree of the Prime Minister of the Republic of Azerbaijan dated June 12, 2023, eight Working Groups were established to ensure the full, timely, and effective implementation of the measures outlined in the "Strategy for Socio-Economic Development of the Republic of Azerbaijan for 2022-2026" and to coordinate the activities of relevant state bodies. The "Innovation" Working Group, within this framework, aims to fulfill the goal of creating a "Creative and Innovative Society" and to coordinate the activities of relevant state bodies.

In 2024, the work of the "Technology and Innovations" Working Group of the Commission on Business Environment and International Ratings has continued. The activities of the working group play a crucial role in improving the country's indicators in the Global Innovation Index. In this direction, necessary work is being done to ensure the continuity of cooperation and communication both within the working group and with other groups and subgroups.

By the Presidential Decree No. 2199 dated July 13, 2016, the Commission on Business Environment and International Ratings was established, and by the Presidential Decree No. 431 dated August 27, 2018, the "Regulation on the Commission on Business Environment and International Ratings" was approved. Currently, within the framework of the Commission, there are a total of 25 working groups, 2 coordination groups, and 4 subgroups on specific areas (environmentally friendly and safe transport, scientific potential, ESG (Environmental, Social, and Corporate Governance) principles, and border trade), consisting of representatives of relevant state bodies and institutions, public legal entities, non-governmental organizations, specialists from enterprises and organizations, and independent experts.

In the first half of 2024, discussions on the draft law of the Republic of Azerbaijan on "Innovation Activity," prepared by the Ministry of Economy, continued. For this purpose,

proposals were received from relevant institutions, discussions were held, and analyses were conducted. Work is ongoing towards the approval of the draft law.

A decision was made, in collaboration with the Ministry of Science and Education, to establish the "Sabah City" Innovation Cluster, and a project has been prepared and presented to the public for this purpose. The innovation cluster "SABAH.city" is planned to be created based on the High Technologies Park of the Ministry of Science and Education. This project will be built on an area of 20 hectares and is expected to open in 2026.

The innovation cluster aims to achieve synergy through the co-location of university campuses, research laboratories, technology transfer centers, and various participants in the innovation ecosystem within a single physical space. "SABAH.city" has the potential to become the epicenter of Azerbaijan's innovation ecosystem by supporting human capital development, job creation, commercialization of scientific research, and knowledge transfer. The project has garnered interest among Azerbaijani youth, and the number of participants is expected to increase.

On June 27-28, 2024, the "Baku ID" innovation festival was held. The event included over 200 local and foreign investors, more than 100 foreign and local startups, 5,000 participants, and over 50 partners. The main sponsors of "Baku ID" were the Ministry of Science and Education, SOCAR, PASHA Holding, ABB, and Mappable, while the partners included 4SIM and IDDA. At the conclusion of the event, a syndicate fund amounting to one million dollars was established with the support of venture funds such as "Startup Wise Guys," "Al-Farabi Innovation Hub," "Caucasus Ventures," "MOST Ventures," "Startupbootcamp," "Aloqa Ventures," and "SABAH.fund."

"Baku ID," one of the main startup innovation events in Azerbaijan and the region, was organized by the SABAH.HUB Innovation Center and brought together startups and investors. This year's "Baku ID" event featured over 200 local and foreign investors, more than 100 startups, and thousands of participants. During the event, winners of the investment fund exceeding \$1,000,000, established with the support of "Startup Wise Guys," "Al-Farabi Innovation Hub," "Caucasus Ventures," "MOST Ventures,"

"Startupbootcamp," and "SABAH.fund," were announced. In total, seven startup teams received investments amounting to \$1,000,000.

The closing event of the grant project titled "Establishment of Innovation Policy in Azerbaijan," implemented since 2022 in collaboration with the Innovation and Digital Development Agency (IDDA) under the Ministry of Digital Development and Transport and the Korea International Cooperation Agency (KOICA), has been held. The project, which began in 2021, aims to create a master plan for innovation for the Innovation and Digital Development Agency, enhance the efficiency of Azerbaijan's innovation policy and implementation process, and strengthen the potential for establishing a sustainable innovation system. To achieve these goals, the Korean Government allocated a grant of \$2,000,000.

As part of the project, startup acceleration (SAP) and training programs for academic staff (TTT) in the field of innovation were organized by the Korea International Cooperation Agency (KOICA) in January-February 2024 on the initiative of the Innovation and Digital Development Agency. During the final "Demo Day" event held on February 6, the graduates of both programs presented their projects.

Participants from various fields were involved in the training, and their final project presentations covered solutions in areas such as the service sector, employment, drone technology, energy, and online shopping. At the end of the event, the top 10 startup teams from the "SAP" program and the top 3 mentors from the "TTT" program were awarded "Samsung Galaxy Tab S9" tablets. In total, 20 startup teams and 20 mentors successfully completed the training programs. All participants of both programs were awarded graduation certificates.

During the "SAP" program, participants acquired essential knowledge and skills such as testing a business idea through hypotheses, conducting interviews with potential customer audiences, advancing the idea to the Minimum Viable Product (MVP) stage, and creating a Business Model Canvas. The "TTT" program was designed for academic staff working on science-intensive innovative projects. In the five-week program, participants improved their

existing competencies by gaining comprehensive knowledge in organizing startup mentorship and preparing training and practical tasks on innovative entrepreneurship.

In 2024, the "Technest" Scholarship Program implemented by the Innovation and Digital Development Agency (IDDA) was continued. The program was conducted in over 50 specialties offered by more than 40 partner educational centers operating in the ICT field across the country, including Baku and several regions.

The "Technest" scholarship program is a national scholarship program providing scholarships in the field of information and communication technologies. The goal is to contribute to the creation of a skilled workforce in the ICT field in the country and to meet the workforce needs in the labor market. In the 2021-2022 academic year, scholarships were awarded to more than 600 individuals, and in the 2022-2023 academic year, more than 3,000 individuals received scholarships. The "Technest" scholarship program, in cooperation with the country's leading educational centers in the ICT field, organizes competitive admission to the most in-demand IT courses to train professionals. The competition is conducted in several stages: online application, one or more exams according to the training program, and finally, an interview stage. Only citizens of the Republic of Azerbaijan aged 13-65 who have successfully passed all stages can benefit from the scholarship.

The "Towards Digital Azerbaijan" project is being implemented with the aim of increasing digital literacy in the regions, developing knowledge and skills in ICT, measuring the existing potential of the region in the ICT sector, taking measures based on continuous monitoring, and supporting development. The "Towards Digital Azerbaijan" project is being implemented with the aim of increasing digital literacy in the regions, developing knowledge and skills in ICT, measuring the existing potential of the region in the ICT sector, taking measures based on continuous monitoring, and supporting development. Within the framework of the regional project, launched by the Innovation and Digital Development Agency on November 24, 2022, it is planned to hold educational and informative training sessions with residents of more than 60 cities and regions of Azerbaijan who have an

interest and experience in ICT, work in this field, study, or aim to build a career in this area. In addition, bilateral meetings with executive bodies, relevant organizations, and institutions will be held. One of the priorities of the "Towards Digital Azerbaijan" project is to train specialists in the ICT sector and ensure their employment in their respective regions. For this purpose, during visits to the regions, training sessions on future innovative professions are being held within the framework of the "Technest" Scholarship Program, with the participation of mentors and leading educational centers of the country, and interactive discussions are organized. Participants have been tested on-site by the educational centers and included in the scholarship program.

The "Digital Skills" project is being continued by the Ministry of Science and Education. The project is conducted to ensure that students deeply acquire ICT skills, achieve independent and purposeful activities in the information space, and develop as competitive individuals with logical and non-standard thinking. The project is implemented by the Ministry of Education and the international educational company "Algoritmika." By improving the teaching of the "Informatics" subject, the project develops algorithmic thinking, logical reasoning, project-building skills in students, and teaches the basics of programming.

The "Creative Discussions" project, presented by the Ministry of Culture, is held in the capital and regions with the aim of discovering new talents in the field of creative industries. Within the framework of the project, more than 100 events have been held in 2024 across 10 clusters. The events are organized in 10 cities and regions, including Baku, Sumgayit, Shamakhi, Sheki, Mingachevir, Ganja, Quba, Lankaran, Nakhchivan, and Shusha. The events consist of master classes, meetings with prominent individuals, and training sessions.

The activities aimed at capacity building under the joint project "Operational Program of the High Technology Park: Implementation and Capacity Building" (2022-2024), implemented by HTP and the Korea Institute of Science and Technology Policy (STEPI), have continued. As part of the work carried out under the project, registration documents have been submitted for registration to establish an "Association of Innovation Ecosystem Players."

A book titled "Innovation and Entrepreneurship: The Startup Ecosystem," prepared in collaboration between the Center for Analysis of Economic Reforms and Communication (CAERC), Baku Engineering University (BMU), and the University of Nevada, USA, has been published. The book compares the role of innovation and startups in ensuring globalization, economic growth, competitiveness, and overall development with models from foreign countries. It also analyzes the institutional issues, current situation, and prospects of the startup ecosystem in Azerbaijan.

An acceleration program titled "Research and Development Manager in Digital Transformation," a joint project of Baku Engineering University's Technopark, "Center of Agile Communication" LLC, and "Yeni Fikir" LLC, has been conducted. The preparation of research and development managers, considered a key driving force of the knowledge economy, including third-generation universities, is becoming increasingly relevant. This acceleration program is implemented by taking into account global, regional, and local market trends. The program, which involves faculty members, consists of both theoretical and practical components and is organized intensively. Additionally, it is planned for each participant to develop their projects individually or in teams during the process. Furthermore, the research and development managers who will be trained will be able to form effective dialogues between universities and industry and prepare projects that offer efficient solutions to businesses.

The "My Technology" project, a joint initiative of Baku Engineering University's Technopark and "Yeni Fikir" LLC, has been conducted. The "My Technology" project carries the mission of promoting and supporting the development of human capital with an entrepreneurial mindset, in line with the demands and expectations of the knowledge economy. The project is organized as a competition, with the participation of 10th-11th grade students from ordinary schools and their teachers, aimed at developing innovative ideas. The "My Technology" competition carries the mission of promoting and supporting the development of human capital with an entrepreneurial mindset, in line with the demands and expectations of the knowledge economy. The project encourages the integration of teachers' and students' innovative initiatives, based on local agricultural specialization,

lifestyle, and global worldview potential, into the innovation ecosystem.

The "Innovations for a Green World" Digital Idea Contest, organized for the third time as part of the "Year of Solidarity for a Green World," received a total of 1,734 applications. Among the registered young participants, the 120 most successful were selected. Gender equality was one of the main principles considered during the selection process.

Registration has started for the "Digital Azerbaijan Talent School" training camp for talented young programmers, organized by the Innovation and Digital Development Agency. The camp is organized in collaboration with "Constructor University," one of Germany's leading universities. The program, which will be held in a hybrid (online and offline) format, will cover the period from August to November 2024. The training camp is based on the requirements of international programming competitions. The training sessions, aimed at developing basic skills in computer science and programming, will be conducted by experts from "Constructor University." In the end, the participants will be awarded international certificates. The camp will be held in two directions: "Advanced Computer Science," for preparation for international programming Olympiads, and "Girls in IT," to promote ICT among girls.

There were interesting activities in the innovation ecosystem in the second half of 2024.

Azerbaijan's innovation ecosystem has been actively advancing, with notable developments in support of startups, digital transformation, and international collaborations. The Innovation and Digital Development Agency (IDDA) has been a major driver, backing initiatives that support both local and international partnerships. For instance, IDDA recently represented Azerbaijani technology at GITEX GLOBAL 2024 in Dubai, a leading technology event, where Azerbaijani startups like PayRiff, GLORRI, and cybersecurity firm CYPHO showcased their products. This participation facilitated connections with global tech leaders, enhancing the country's international presence in fields like fintech and cybersecurity.

Another milestone for Azerbaijan's innovation landscape was the opening of the Azerbaijan Innovation Center (AIC) in October 2024. Designed to support startups and foster innovation,

AIC offers incubation, acceleration programs, and networking opportunities for entrepreneurs. The center is expected to expand its reach to other cities, making resources more accessible nationwide. This aligns with the government's broader goal to stimulate regional innovation hubs, creating a fertile environment for startups and investors alike.

UNIDO has also been instrumental in bolstering Azerbaijan's capacity for the Fourth Industrial Revolution (4IR). Collaborating with national stakeholders, UNIDO has provided training in innovation management and digital transformation, with a focus on helping Azerbaijani businesses adapt to global standards, especially within EU markets. This partnership is expected to facilitate the application of advanced technologies across multiple sectors, fostering Azerbaijan's long-term digital transformation goals.

These initiatives underscore Azerbaijan's commitment to building a robust, globally integrated innovation ecosystem through strategic investments in technology, partnerships, and infrastructure, positioning the country as an emerging hub for digital solutions and entrepreneurship in the region.

The Azerbaijan Cybersecurity Center, operating under the Innovation and Digital Development Agency (IRIA), has organized "Purple Team" training sessions to enhance the knowledge and skills of cybersecurity professionals from both the private and public sectors. Notably, this is the first time such training is being conducted in this field in our country.

The "Purple Team" training sessions are designed for individuals with a minimum of two years of experience in cybersecurity. The lessons are taught by international experts and provide an excellent opportunity for specialists who are focused on either "red team" (attack) or "blue team" (defense) techniques.

The purpose of these training sessions is to develop the highest level of knowledge and skills among the country's cybersecurity personnel.

The training program for this cohort will span one month. The training sessions are expected to be ongoing and will eventually be extended to include cybersecurity professionals from other sectors as well.

On August 2, 2024, the graduation ceremony of the "IDDA Academy" internship program, organized by the Innovation and Digital Development Agency (IDDA), took place. Out of 721 applicants, 38 individuals were selected for the program, and 25 of them successfully completed it. In her opening speech, the Chairwoman of the Agency, Inara Valiyeva, expressed her gratitude to the employees who supported the interns with their knowledge and mentorship throughout the program. She thanked the graduates for their participation in the "IDDA Academy" program, congratulated them on their achievements, and wished them success in their future careers. At the end of the event, all interns were awarded participation certificates, while those with outstanding results received distinction certificates.

Notably, during the three-month-long "IDDA Academy" internship program, interns gained both theoretical and practical skills by working on real projects with agency staff in areas such as cybersecurity, business solutions, data analytics, and various ICT fields. The program included several initiatives to ensure the interns' initial orientation and rapid adaptation, such as an "Orientation Day," an intensive training program to develop various skills, debate training, and programs like "Digitalks" and "Defence Day."

For the first time, Azerbaijan is hosting the regional event of the "Startup World Cup", considered the most prestigious world championship for startups. On September 19, the regional stage of the Startup World Championship was organised for the first time in our country by the Technovate-SABAH Angel Investors Club at ADA University. The competition, regarded as one of the most prestigious startup competitions in the global arena, is of great significance in contributing to the development of the startup and investment ecosystem in Azerbaijan by bringing together investors, company leaders, startup founders, and various government officials.

Within the competition's framework, local startups will have the opportunity to present their entrepreneurial ideas on an international stage. A seven-member jury, representing various sectors, will evaluate the projects. Ultimately, the winning startup will represent Azerbaijan in the final stage of the competition in the USA.

The following activities have been carried out in the direction of technology transfer and commercialization of science, utilizing Korea's experience in this field:

"Innovation Ecosystem Players Association" -Under the joint project of the Ministry of Science and Education's High Technologies Park (HTP) and the Korean Institute of Science and Technology Policy (STEPI), titled "High Technologies Park Operation Program: Application and Capacity Building" (2022-2024), capacity-building activities have continued. As part of the project, documents for establishing the "Innovation Ecosystem Players Association" were prepared and submitted for registration.

"Research and Development Manager in Digital Transformation"- An acceleration program titled "Research and Development Manager in Digital Transformation" was conducted as a joint project of the BEU Technopark, Center of Agile Communication LLC, and Yeni Fikir LLC. Over 20 instructors successfully completed training in the "Digital Product Mindset and Competencies" module. The program, focused on developing research and development managers, a driving force of the knowledge economy and third-generation universities, addresses global, regional, and local market trends. This intensive program involves theoretical and practical components, including the development of individual or team projects by participants. The trained managers are expected to foster effective dialogue between universities and industry and to design projects providing effective solutions for businesses. Additional modules in the program are ongoing.

<https://beu.edu.az/az/article/bmu-da-tedqiqat-ve-inkisaf-menecerlerinin-hazirlanmasi-ile-bagli-telimler-basa-catib-2176>

Incubation Program is a 3-month incubation program (April–June 2024) was organized at the BEU Technopark, benefiting 12 startups. Throughout the program, startups received specialized training, mentorship, and other services. The program concluded with an exhibition and demo day, where the startups graduated.

<https://beu.edu.az/az/article/new-startups-of-beu-students-presented-2039>

"My Technology" is a joint project of BEU Technopark and Yeni Fikir LLC, was sponsored by Kapital Bank. This project supports and promotes human capital with entrepreneurial

thinking in line with the demands and expectations of the knowledge economy. Organized as a competition, the project engaged 10th and 11th grade students and their teachers from secondary schools to develop innovative ideas. The project fosters the integration of teachers' and students' local specialization and lifestyle perspectives with a global worldview into the innovation ecosystem. Over 200 teachers and students received training and mentorship, and more than 60 projects were developed, with 35 of them showcased as MVPs and other formats. <https://www.facebook.com/mytechnology.az>

"IT Project Management and Startup Skills" The "IT Project Management and Startup Skills" project, a collaboration between Yeni Fikir LLC and BEU, was organized under the Technest scholarship program by the Innovation and Digital Development Agency. Held from April to July 2024, the project aimed to develop the IT project management and startup skills of over 20 BEU students. With professional training from Anar Rustamov and Isa Qasimov, participants applied the acquired knowledge and skills in startup projects, enhancing their practical experience. The training sessions were conducted at BEU Technopark. <https://yenifikir.az/it-project-management-startup-skills-layih%c9%99si-ugurla-basa-catdi/>

"Academic Entrepreneurship-Based Startup and Incubation Program" The "Academic Entrepreneurship-Based Startup and Incubation Program" (ASSIP) project aims to train teacher-mentors in "academic entrepreneurship" skills, integrate them into the innovation ecosystem, and facilitate the launch of new startups in the market with their involvement. The project includes two rounds of training-of-trainers (TOT), two hackathons, and two incubation programs. To date, one TOT has been completed, with 35 academic staff members participating. Registration for the first hackathon is ongoing. The ASSIP project is held at BEU with the support of the Innovation and Digital Development Agency and is implemented by BEU Technopark and Yeni Fikir LLC.

<https://www.facebook.com/groups/1200256157934641>

Professor Isa Qasimov and Bunyamin Seyidov have published informative articles in the media:

- "The University and Its Technopark are Like 'Zero' and 'Phase'" Isa Gasimov
<https://valyuta.az/az/ekspert/48897/universitet-ve-onun-texnoparki-nol-ve-faza-kimidir-biri-olmayanda-isiq-yanmir/>
- "Is It About Graduates Finding 'Good Jobs' or About How Many Companies Emerge from the University?" Isa Gasimov
<https://valyuta.az/az/ekspert/49779/mezunlarin-yaxsi-is-tapmasi-yoxsa-universitetden-nece-sirket-ortaya-cixmasi/>
- "The Social Impact of Cyber Threats" B.Seyidov

3. Micro-Level Analysis of Operational and Managerial Issues at the HTP

SABAH.city is a transformative innovation cluster slated for launch in 2026, driven by the Ministry of Science and Education in collaboration with key public and private stakeholders. The initiative aims to create a thriving ecosystem that fosters research, development, and innovation, positioning SABAH.city as a major regional hub. The development is planned across three strategic phases:

- **Phase 1: 2024-2026 - Laying the foundations and building initial traction**

During this foundational phase, SABAH.city will begin establishing initial facilities across approximately 4 hectares, setting the stage for future growth. Key components, including startup incubators, accelerators, venture studios, venture capital funds, and technology transfer offices, will be operational, providing essential support for emerging businesses. In addition, leading public and private educational institutions and pioneering companies will relocate and establish themselves as the initial residents of SABAH.city, contributing to a vibrant community focused on innovation.

- **Phase 2: 2026-2029 - Scaling up and expanding capabilities**

In the second phase, the developed area will expand to roughly 15 hectares, enabling the construction of additional infrastructure and enhanced facilities. This phase will introduce cutting-edge research centers, state-of-the-art data centers, and advanced co-working spaces designed to foster collaboration and drive innovation. By this stage, SABAH.city will further integrate leading educational institutions and innovative companies as foundational residents, cementing its role as a dynamic environment for research and technological advancement.

- **Phase 3: 2029 and Beyond - further expansion and internationalization**

In the final phase, SABAHA.city will extend across the full 20 hectares, maximizing resource utilization and spatial efficiency. This expansion will include welcoming leading international companies seeking to establish regional offices, thereby expanding the city's global reach. With this growth, SABAHA.city will solidify its position as a premier hub for research, development, and innovation, attracting talent and investment from across the globe.

SABAHA.city promises to be a landmark destination for innovation, bringing together academia, industry, and investors in a collaborative environment that supports sustainable growth and technological breakthroughs.

SABAHA.city provides a comprehensive ecosystem designed to support innovation and growth for startups and technology companies. Their offerings include:

- Access to talent: Connecting businesses with skilled professionals across various fields.
- Media coverage and PR support: Helping startups gain visibility and promote their stories.
- Networking with tech companies and startups: Fostering connections within the tech community for partnerships and growth opportunities.
- Startup and tech academies: Educational programs to develop essential skills for the startup and tech industries.
- Technology transfer programs: Facilitating the transfer of cutting-edge technology from research to market.
- Venture studios: Providing resources and expertise to develop new businesses from the ground up.
- Incubation and acceleration programs: Offering structured support for early-stage startups to scale and succeed.
- Offices and co-working spaces: Flexible workspace options designed to foster collaboration.

- Sharable research labs: Access to high-quality labs equipped for research and development.
- Tax incentives: Financial benefits to support startup growth.
- Pre-seed & seed financing and venture funds: Funding opportunities for startups at various stages.
- Corporate financing options: Financial support for established tech businesses looking to expand.

SABAH.city brings together resources, expertise, and a collaborative environment to accelerate innovation and support the entire lifecycle of tech startups and companies.

Activities in the framework of the project- “Operational Program of the High Technology Park: Implementation and capacity building”

On February 28, 2024, a kickoff meeting was held as part of the project “Operational Program of the High Technology Park: Implementation and capacity building” The meeting was attended, Korean and local experts and specialists.

The project coordinator Dr. Eun Joo Kim opened the event with an introductory speech, providing general information about the work carried out within the project and informing participants about the support provided by Korea.

In his speech, Vasif Abbasov, Director General of the High-Tech Park, discussed the collaboration between the High-Tech Park and STEPI, highlighting the considerable progress made over the past period. He emphasized the significance of the project for Azerbaijani beneficiaries and expressed high expectations for the current year’s activities, stating that the High-Tech Park would always support the project. He also mentioned that memorandums had been signed with other Korean organizations and agreements had been reached on practical work.

Subsequently, presentations were made by local and Korean experts on the state of technology commercialization systems in Azerbaijan, work conducted within the framework

of “K-Innovation 2023,” the improvement of the Technology Transfer and Commercialization Platform, and the training of Azerbaijani trainers under Korea’s I-Corps program, followed by discussions on these topics.

As part of the project “Operational Program of the High Technology Park: Implementation and capacity building,” a 4-week seminar was organized. During the seminar on technology transfer value, lectures were held on the in-depth study of Korea's technology transfer practices for trainers. Participants also expressed their views on the topics based on assignments given after the lectures. The lectures were delivered by Seung Ho Lee, CEO of “Delta Tech Korea” and a renowned expert in technological innovation management in Korea. The seminar was attended by active members of the innovation ecosystem, including representatives from Baku Engineering University, ADA University, Baku Higher Oil School, Architecture and Construction University, High-Tech Park, Western Caspian University, Azerconnect, Odar Yurdu University, and the Institute of Management Systems.

The main objective of the seminar was to teach participants to analyze successful and unsuccessful cases based on previous experiences, manage virtual organizations, and create value through human resources. Participants who successfully completed the seminar were awarded certificates.

In 2024, a training program for trainers was held as part of the project “Operational Program of the High Technolgy Park: Implementation and capacity building” The training program is part of Korea's “Korean I-Corps Program” and “Lean Startup & Customer Discovery” programs, and its curriculum is based on the U.S. National Science Foundation’s “I-Corps” program. The training program provides comprehensive knowledge on business model components and customer discovery for startups. The program will continue in traditional format in Korea, online format, and finally, in October, with a forum on technology commercialization in Baku. The program aims to enhance Azerbaijani experts' knowledge in the mentioned areas and to develop startups in the country through experience and knowledge sharing.

On the first day of the training, Nihat Majidov, Director General of the High-Tech Park of the Ministry of Science and Education, spoke about the work done in collaboration with Korea's STEPI and highlighted that this year's training would be beneficial for both the HighTechnology Park and other higher education institutions and innovation centers in terms of skill enhancement.

The Director General also discussed future developments of the High Technology Park, mentioning that the new "Sabah City" innovation cluster to be established at the technopark is planned to be a venue where innovation processes in Azerbaijan will flourish, technology transfer and commercialization will be implemented, and startups can further develop themselves.

The presentations for the first day of the training were then conducted. Professor John Shim from Seoul National University of Science and Technology provided detailed information about the "I-Corps" program and an extensive explanation of the training process until October.

Roy Hvang, Executive Director of Korea's "INSIGHTORS" company, introduced a broad presentation on business models and value propositions. Participants engaged with theoretical and practical knowledge during the presentation, which included assignments.

On the second day of the training, Roy Hvang presented on Business Model Canvas and Business Model Evaluation. Six participants who successfully participated in these training sessions were granted the opportunity to attend the traditional format training sessions in Korea.

Additionally, a report on the activities of 2023 was translated into Azerbaijani and prepared for publication. The published books are planned to be distributed to universities, innovation centers, and other organizations.

■ Activities Planned for the second half of 2024

The “Operational Program of the High Technology Park: Implementation and capacity building” project, prepared by local experts, includes the implementation of the "Academic Entrepreneurship-Based Startup and Incubation Program." The program's organization is expected to incorporate experiences gained from the Korean experience, specifically from Korea's "Korean I-Corps Program," "Lean Startup & Customer Discovery" programs.

The "Operation Program of the High Technology Park: Implementation and capacity building" project, prepared by local experts, includes the organization of the "My Technology Azerbaijan" prototype competition. The "My Technology Azerbaijan" prototype competition will be implemented by TTCP (Technology Transfer and Commercialization Program).

If the "Innovation Ecosystem Players Association" is officially registered, it is planned to hold extensive events jointly with TTCP.

Acceleration of activities related to TTCP, enrichment of content, and further intensification of activities involving data collection are planned. Additionally, information days are planned to be held at various universities.

Dissemination days with the participation of local trainers are planned (in September-October) to spread the knowledge acquired through the training programs held in Korea.

A National Forum on Technology Commercialization is planned for October, with the participation of Korean and Azerbaijani experts and leading participants of the innovation ecosystem.

As part of the project "Operational Program of the High Technologies Park: Implementation and Capacity Building," a series of training sessions and meetings were held in Korea.

From August 25 to 29, 2024, a training seminar for startup instructors took place in Seoul, the capital of the Republic of Korea, under the "2024 K-Innovation Partnership Program with Azerbaijan." The seminar was attended by Bunyamin Seyidov, Head of the Incubation and

Technology Transfer Center at the High Technology Park; Associate Professor Isa Qasimov, Director of the Baku Engineering University Technopark; Associate Professor Muhammedfoik Bahadyrov of ADA University; Gunel Imamquliyeva, Chairperson of the Student Scientific Society at the Azerbaijan University of Architecture and Construction; Ravana Goyushlu, Head of the Startup and Incubation Center at Odar Yurdu University; and Nermin Ismayilova, a startup specialist at the Baku Engineering University Technopark.

The primary goal of the training was to enhance the skills of selected startup instructors from Azerbaijan in business model development and provide them with practical experience in Korea's startup support program. During the sessions, experts from Seoul National University, Seoul National University of Science and Technology, Delta Tech Korea, INSIGHTORS, and the Korea Institute of Science and Technology Policy (STEPI) delivered lectures, focusing on the exchange of international technology startup practices and exploring Korea's I-Corp model. Additionally, the six Azerbaijani specialists presented their projects on business model development for startups in two groups, which were positively evaluated by the Korean experts. At the end of the two-day training, the participants were awarded certificates. The Azerbaijani specialists also participated in video interviews discussing the implementation, outcomes, and future cooperation plans related to the project.

Moreover, the Korean partners were introduced to the SABAH.city innovation cluster, set to be established within the High Technology Park, and potential collaboration opportunities with relevant Korean organizations were discussed. The Azerbaijani startup specialists also visited HABSIDA in Incheon to examine examples of international technology startups and met with the company's CEO, Vladi Mun. Additionally, they attended a presentation by Insil Baek, the head of Kakao Mobility, to learn from the successful experiences of Korea's I-Corp program.

During their visit, the delegation also visited the Embassy of the Republic of Azerbaijan in Seoul and met with Ambassador Ramin Hasanov. During this meeting, they discussed the current state of Azerbaijan's innovation ecosystem and issues related to startup

development. Ambassador Hasanov expressed the embassy's readiness to provide necessary support in these areas, including facilitating coordination with relevant Korean organizations.

In September and October 2024 three-time office hour program launched to bolster practical education and training in Azerbaijan. The "Three-Time Office Hour" initiative has been launched to enhance the practical skills and training of prospective instructors in Azerbaijan, focusing on customer discovery-based experiences. This program is designed to build the capabilities of instructors through a series of structured sessions aimed at deepening understanding of target technologies and developing effective business model presentations.

To meet program expectations, participants are required to conduct a minimum of 10 customer discovery interviews per session. In the first session, participants are encouraged to interview an inventor to gain a comprehensive understanding of the target technology. Additionally, the Entrepreneur Leader (EL) should lead the team presentation whenever possible, contributing to an immersive experience in customer-focused discussions.

All three Office Hour sessions are mandatory, with participants expected to use a designated template for their presentation materials.

The program also includes individual tasks for each prospective instructor. During the first session, after team presentations, a special lecture is scheduled to provide additional insights. For the second and third sessions, instructors must select one of the nine business model blocks, develop a unique lecture based on their choice, and deliver a 20-minute mini-lecture to Korean instructors. Through this initiative, the STEPI K-Innovation program seeks to foster a dynamic, real-world learning environment for Azerbaijan's prospective instructors, equipping them with essential customer discovery skills and business model acumen.

With support from the Korean government and partnership with the Science and Technology Policy Institute (STEPI), a delegation from Korea visited Azerbaijan from October 21-25 as part of the "High-Tech Park Operations Program: Application and Skills

Enhancement" project, running from 2022 to 2024.

During the visit, a meeting took place on October 22 with Nihat Majidov, CEO of the High-Tech Park, resulting in a Memorandum of Understanding signed with DeltaTech.

On October 23, a final seminar on "Commercializing Technologies" was held, where Korean experts presented topics in various projects and local experts shared detailed insights into the project's activities and experiences.

As part of the visit, the delegation also held meetings with the State Agency for Science and Higher Education, Azerbaijan University of Architecture and Construction, Baku Engineering University, SABA.HUB, and the Coders office.

At a meeting held at Baku Engineering University (BMU), BMU's Vice-Rector for International Relations, Atrabe Gul, emphasized the successful development of bilateral relations and highlighted the positive aspects of cooperation in higher education. During the meeting, Mayis Azizov, Head of Department at BMU, provided information about the university's science and innovation-focused activities. He also mentioned the wide opportunities available for collaboration in conducting joint research in information technology.

Isa Gasimov, Director of the BMU Technopark and one of Azerbaijan's experts on the project, recalled previous visits by the STEPI delegation to Azerbaijan. He noted that the established cooperation was aimed at achieving positive outcomes in technology transfer and the commercialization of scientific research.

Bunyamin Seyidov, Head of HTP, also participated in the event and discussed the project, underscoring the importance of Azerbaijan-Korea scientific and technical relations.

The Korean representatives stated that STEPI supports the development of an innovation ecosystem in Azerbaijan and expressed their continued readiness to assist in this effort. They also provided information on the implementation strategies and current outcomes of the technology transfer project.

In the discussions, the potential for cooperation between BMU and STEPI was explored, including the development of joint programs, expansion of collaborative initiatives, and commercialization of technological ideas and startups by BMU's faculty and students. Representatives from the ASSiP (Academic Entrepreneurship-Based Start-Up and Incubation Program) project, implemented by the BMU Technopark, also presented their proposals, and received a number of recommendations from Korean experts for advancing their academic startups.

During the Korean delegation visit a meeting was held at the Azerbaijan University of Architecture and Construction (AzMIU) with a delegation from Korea. The meeting was organized by the High Technologies Park under the Ministry of Science and Education.

AzMIU's rector, Professor Gulchohra Mammadova, welcomed the Korean guests, provided information about the university, and discussed prospects for joint cooperation.

The meeting was attended by Seong Ho Lee, the head of Deltatech - KOREA; Professor John Shim from Seoul National University of Science and Technology; Professor Youngtack Shim from Seoul National University; and Bunyamin Seyidov, the head of the Department of Incubation and Technology Transfer at the High Technologies Park.

The parties discussed technological solutions achieved in Korea's architecture and construction sector and the possibilities for AzMIU to benefit from these solutions.

4. Conclusion

This Report on the Status of the K-Innovation Partnership Program 2022-2024 between STEPI and the HTP can be considered a follow-up to the previous status report submitted in 2022. This report mainly concentrates on the updates that have occurred since July 2022, while considerable institutional reforms have been implemented within ANAS. Numerous institutions, mostly in the field of natural and technical sciences, have been transferred within the structure of the Ministry of Science and Education. As for ANAS, it mostly manages institutes in humanitarian and social sciences. Most importantly for this project, the ANAS HTP is no longer a unit of ANAS, although the interagency transferring process is still ongoing and needs to be finalized.

Another significant event entails strengthening the authority of the Ministry of Transport and Digital Development as a central power to govern the innovation ecosystem, digital transformation, and e-governance. All these responsibilities are currently assigned to the Innovation and Digital Development Agency. The HTP would benefit from closer collaboration with the IDDA and active participation in the innovation ecosystem.

So far, Azerbaijan has not introduced any legal acts on competition, venture capital, etc. Moreover, a draft of the national innovation strategy needs to be approved in order not to miss the right timing or lose out on the market potential. Equity and debt financing, and crowdfunding are underperforming forms of funding mechanisms in the country. Furthermore, the unavailability of venture capital firms in the domestic market makes the creation of more dynamic markets and financing an even more challenging task.

We believe that the process of re-formation of HTP will take place after it becomes part of the Ministry of Science and Education. The visits of the minister and other management staff to the HTP, and their respective ideas, indicate that the HTP will once again be formed into a modern technology park.

Regarding the need to improve the HTP's operational and managerial issues, it will soon conduct a needs assessment and develop improvement plans based on the findings of the assessment and its own strategic aspirations. Deploying a solid business model will surely have a positive impact on the sustainability and profitability of its operations. Moreover, it is important for the HTP to strengthen its team capacity, monitoring and evaluation system, and participate in the innovation and TC ecosystem more actively.

The K-Innovation Partnership Program is a valuable opportunity for Azerbaijan, and for the HTP in particular, to develop a far better and more conducive environment for technology commercialization.



CHAPTER 3

Policy Recommendations

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Chapter 3. Policy Recommendations

1. Background of the Azerbaijan Model

1.1 Current Status of the Public Sector Start-up Ecosystem in Azerbaijan

1.1.1 Azerbaijan Start up Ecosystem through Preliminary Research

Azerbaijan has a population of approximately 10 million, with around 2 million (20%) residing in Baku. While small-scale B2C businesses are feasible, ICT-based service platforms typically require over 100,000 subscribers to secure more than 1,000 loyal customers who visit at least once a week, indicating a relatively small market size.

Most start-up and incubator support programs are government-led. Although the overall budget plan has been reported, specific data on the investment amounts in various sectors is not available. Representative projects include programs from the following institutions: According to KOTRA (2018), “Azerbaijan Start-up Programs and Promising Fields,” by Myung-Hoon Oh, Baku Trade Office, December 14, 2018, the local government announced the Strategic Road Maps for the National Economy and Main Economic Sectors in December 2016, focusing on the development of non-oil industries and the fostering of SMEs.

In December 2017, the Agency for Small and Medium Business Development of Azerbaijan was established to support SMEs. The ASAN Center, an administrative organization similar to a district office in Korea, provides business centers for SMEs, offering education, consulting, financial, and administrative services. In regions without ASAN Centers, business centers are operated by regional departments under the Ministry of Economy.

In August 2018, the Entrepreneurship Development Fund, previously under the Ministry of Economy, was made independent and allocated a budget of 1.1 billion Azerbaijani Manat (approximately \$645 million) to promote the development of private companies in non-oil industries. The fund supports various activities, from entrepreneur education to project funding, with priority industries being agriculture, tourism, information and communication, and manufacturing.

In March 2018, the Youth Fund was established to provide financial support for start-ups, and tax laws were amended to exempt companies with a Start-up Certificate from income tax for three years. Additionally, the Ministry of Transport, Communications, and High Technologies, in collaboration with the Heydar Aliyev Foundation, hosts start-up competitions.

Representative start-up and incubator support programs include the following institutions and infrastructures: Start-up Azerbaijan: Established by the Ministry of Transport, Communications, and High Technologies to encourage start-up creation and strengthen connections with new start-ups. Website: www.start-up.az

Azerbaijan Start-up EcoSystem: A site that gathers all information about the start-up ecosystem for young entrepreneurs. Website: www.start-upazerbaijan.az

Next Step Innovation Center: Supports start-up founders from product concept to prototype development, providing funding to those who complete the center's education programs. Office space is provided in Baku for selected prospective entrepreneurs. Website: www.icnextstep.com

Social Innovation Lab (SIL): A research institute established to nurture 50 tech start-ups annually by 2025, offering entrepreneur education and overseas market entry services. Website: www.socialinnovationlab.az

SUP: An intensive education program for start-up nurturing, offering a three-month course on product invention, marketing practices, and more. Website: www.sup.az

Local telecom start-up support programs:

Barama: A start-up support program by telecom operator Azercell Telecom. Website: www.barama.az

Bakcell AppLab: A program by telecom operator Bakcell to support mobile app development by start-up founders. Website: www.applab.bakcell.com

University start-up programs: Start-up support centers at Azerbaijan State Oil and Industry University and ADA University.

However, some Korean experts have a somewhat negative view of the overall evaluation of local start-ups and support programs in Azerbaijan.

Despite government-led start-up and incubator programs, the poor business environment often results in these programs being run as showcases rather than effective support mechanisms.

The number of start-ups created through these programs is very low. The following are representative local conditions that hinder start-up development:

- (Closed Economy) Due to its geographical proximity to Russia and Turkey and its small market size, there is little foreign business entry. The private sector lacks capital liquidity, and a closed economic structure with capital concentrated in a few companies prevails
- (Brain Drain) The dominance of vested interests and a society based on blood and regional ties make it difficult for those without connections to join the elite, leading to a significant outflow of talented individuals abroad
- (Large Corporation-Centric Economy) The implicit acceptance of high-ranking officials owning companies results in large corporations owned by high-ranking officials dominating industries, making it difficult for new start-ups to emerge
- (Tax Burden) Frequent tax audits and high taxation on private companies lead to many businesses operating informally

- (Education System) Low salaries for teachers and poor educational facilities result in generally low-quality education

1.1.2 Results of the Local Start-up Ecosystem Survey in Azerbaijan through K-Innovation

Over the course of 2022 and 2023, a local market survey was conducted with major universities and start-up support institutions, and the main interview results are as follows:

In Azerbaijan, there are no dedicated funds to support start-ups within universities, nor are there significant government R&D funds to promote research and development, which is the seed of entrepreneurship.

Some universities have been promoting entrepreneurship through education in the information and communication sector, which requires relatively low initial capital for start-ups, since the mid-2010s. However, due to the lack of follow-up investment or support funds after completing entrepreneurship education, it is difficult to achieve substantial start-up incubation beyond operating educational programs.

Major universities in Azerbaijan generally operate start-up education at the undergraduate level and hope to create a start-up ecosystem considering the industrial characteristics of Azerbaijan. However, they face difficulties in raising funds to support this, highlighting the need to establish a start-up incubation system linked with overseas accelerators or venture capital from countries like the United States.

Unlike major countries such as the United States, Korea, and Europe, which preemptively hold intellectual property and patent pools to protect and differentiate start-up business items, Azerbaijan still perceives securing intellectual property through research and development in science and technology as a public good, resulting in a lack of motivation to actively secure intellectual property.

Additionally, although some universities in Azerbaijan file patents, it is difficult to know which patents have been filed between universities, making it challenging to obtain

accurate statistics on the number of patents filed by universities.

It is estimated that there are approximately 200 start-ups certified as “Kovika” (similar to Korea’s venture certification) in Azerbaijan, and including uncertified companies, there are about 300 to a maximum of 400 start-ups.

When asked about promising start-up fields, agriculture, tourism, information and communication, and manufacturing were mentioned, with some interviewees highlighting the medical field. This is because medical expenses of about USD \$150M per year are incurred in countries such as Turkey, Iran, Germany, and Korea, indicating the potential for deep-tech companies to enter the medical-related market. Although medical education is well conducted in Azerbaijan, practical training is weak, and doctors’ salaries are not very high, so improvements in these areas are also needed.

Regarding the creation of an investment ecosystem, unless there are business items targeting a market size of at least USD \$30M to USD \$100M, aggressive inflows of venture capital will not be easy. In this regard, ICT is considered the most fundamental technology field, and promising business models include ICT platform services, smart farms, cosmetics using oil-related by-products, second-hand trade-related industries, and franchise business models reflecting the characteristics of neighboring countries.

Currently, the external appearance of Azerbaijan’s start-up investment ecosystem is publicly led, but in reality, it is privately led, with university-based start-ups or patent-based technology start-ups being almost non-existent or in the early stages.

1.1.2.1 Local Start-up Support Institutions and Organizations

The results of local interviews with major universities and start-up support institutions in Azerbaijan are as follows:

(Innovation & Digital Agency) Established with the aim of transforming Azerbaijan into a technology-based entrepreneurial nation and creating a digital government. The organization, primarily focused on innovation ecosystems and digital development, aims to:

Build an innovation ecosystem by nurturing talent, developing tech companies, supporting the investment ecosystem, spreading entrepreneurship, and establishing infrastructure like Absheron Valley (a cluster of shared offices and related infrastructure).

Promote digital development by creating a digital government, society, infrastructure, business, and ecosystem, collaborating with various countries.

The agency's five main areas of activity are:

Supporting the investment ecosystem, including business angels clubs support, matching fund development, and VC funds creation support.

Developing technical human resources through skills enhancement, peer-to-peer tech skills education, entrepreneurship education, investor education, media education, STEM Olympiads, internships and job matching between investors and ecosystem members, mentorship programs, and innovation champion education.

Focusing on tech company development, including university innovation support, corporate innovation development, program incubation, acceleration programs, and venture studio support.

Promoting the tech ecosystem through ecosystem matching, organizational roadshows, branding, positioning, tech media support, vertical community support, event and networking support, and IDDA award support.

Developing Absheron Valley & improving legislation, focusing on innovation space development, tech ecosystem composition, technopark creation support, and relocation program development & support.

(SABAH.CITY and HTP) In the design phase of Sabah City, various global technology clusters such as Turkey's technoparks, Uzbekistan's IT Park, Silicon Valley in Russia and the USA, Indonesia, and Korea's technoparks were benchmarked. Among these, the main benchmarking target was the successful example of Turkey's technopark established by Turkey's top universities. Due to the small market size, the project considers market

innovation and startup growth entry by integrating markets with Central Asian countries (Uzbekistan, Kyrgyzstan, etc.). The Sabah City project has four main goals: (1) Cultivating talented individuals by attracting universities, (2) Research and development for commercialization, (3) Industry-university cooperation related to knowledge/technology transfer, (4) Job creation through the establishment of a knowledge-based economy. Sabah City is expected to absorb and operate public-led innovation funds like Sabah Hub, and various programs will be operated, making it one of the three largest infrastructures in Baku, the capital. For resident companies, soft support such as tax exemptions (20%) and regulatory innovations are being considered. Under the active support of the government, six major Azerbaijani companies, including SOCAR, ABB, and PASHA Holding, have decided to participate. The project will be carried out in three phases: the first phase from 2024 to 2026, the second phase from 2027 to 2029, and the final completion in 2030. Regarding employment visas, cooperation with economic departments is ongoing.

(State Agency of Science and Higher Education) The agency plays a key role in the construction of Sabah City. It acts as a direct advisory body under the Ministry of Education, supporting various stakeholders in scientific and technological fields, including HTP, and overseeing overall planning. Currently, Azerbaijan shows interest and plans for investment in commercializing public technology developed through research and development. However, there are limitations in the quantity of existing patents and the institutional framework, which pose challenges in fostering high-level innovative startups. Therefore, the government considers providing support and improving systems, focusing on approximately 18 universities under its jurisdiction.

(Sabah Hub) Launched around 2021 under the leadership of the Ministry of Science and Technology of Azerbaijan. Initially started as an acceleration program, it operates various programs, including educational ones, in collaboration with Europe's largest business school, IE Business School. After IE Business School, the collaboration in education expanded by utilizing experts from Stanford and MIT, focusing mainly on entrepreneurship, peer-to-peer IT-related education, design education, and business education. For investment attraction through IR pitching, there are events like Baku Investment Day. Initially, there were about

1,000 participants, 50 VCs, and 100 startups, of which 30 were foreign companies. However, as the startup culture in Azerbaijan is still in its early stages, the purpose of establishing the Sabah incubation organization is to nurture talent through young entrepreneur education, create infrastructure, and enhance fund support. The target fund amount is approximately USD \$25M, but since the size of government funds is currently not large, it relies on private companies like PASHA, SOCAR, etc. Sabah Hub and Sabah Lab are mainly operated, with a deep partnership with IE Business School (university). Startup education is conducted through Sabah Academy, by finding professors abroad (especially in Europe). A leading company in the healthcare field is one of the representative startups and is one of the fastest-growing startups in Azerbaijan. As one of the organizations leading the establishment of an innovative system in Azerbaijan, it aims to contribute to economic development through the spread of entrepreneurship. The organization's major activities are centered around three directions: (1) talent cultivation and human capital development, (2) infrastructure support, and (3) providing connections with capital. Human capital development is carried out through Sabah Academy (mainly operated with resources from universities, with programs autonomously created by the universities, educating about 1.5 million people annually, a threefold increase compared to the launch year), Peer Stack (peer-to-peer technology education), and Sabah School (focused on K-12 to develop future innovative entrepreneurs' capabilities). Infrastructure support includes Sabah Lab and BakulD (an event that matches investors with startups, including those from various neighboring countries, with companies receiving investments averaging about \$15,000 and totaling about \$3 million as of 2024), and the Sabah City project (linking all stakeholders in the innovation ecosystem into infrastructure). The capital linkage is facilitated through Sabah Studio and Sabah Fund, aiming for detailed performance connections. Each activity encourages company participation, with human capital development involving NEQSOL Holdings, Landau School, infrastructure support from SOCAR (Azerbaijan's largest company) and PASHA Holdings, and capital involvement from SAPIA Holdings, with public government funds linked to program operations. Just 1-2 years ago, it was not believed that the current achievement of nurturing about 80-100 startups by Sabah Hub was possible. Similarly, for BakulD (focused on investment in five fields: education (main), health, green, logistics, and

agriculture), few believed that such investment events would be possible in Azerbaijan until 2023, but now encouraging changes with tangible achievements are emerging in these areas. One issue is that the current age of startup entrepreneurs is very young, and there are few experienced professionals with diverse industry backgrounds, which could pose a problem in areas such as mentoring.

(INNOLAND) Moved to its current facility in 2018. Operates similarly to Korea's start-up-supporting shared offices, with most resident start-ups in the ICT sector. There are no companies with separate research labs, and it operates 24/7 with government support. An accelerator supporting seed funds is stationed within the facility, capable of pre-seed stage fund investments.

(Investment Ecosystem) The representative start-up investment fund in Azerbaijan is the Sabah Fund, which includes joint funds with private entities and participation from the Asian Development Bank and the European Development Bank. However, the Sabah Fund targets only companies residing in Sabah Hub. The Sabah Fund can also invest in start-up entrepreneurs in Kazakhstan, Estonia, Uzbekistan, and Turkey. The exact number of venture capital firms in Baku is unknown, but includes private entities like Caucasus Ventures, and some from Estonia, Kazakhstan, and the US. The scale of these start-up investment funds is still small, so they are generally seen as cooperative rather than competitive.

(Others) In Azerbaijan, there are many start-up incubating spaces similar to Korea's shared offices. It is necessary to strengthen financial support for tech-based start-ups residing in these facilities to create a tech-innovation-oriented start-up ecosystem centered around major metropolitan areas and specific regions.

1.1.2.2 University Start-up Support Programs

(Baku State University) Baku State University has an on-campus innovation center that encourages research based on science and technology. However, the number of research and development projects is not high, and there is not much awareness of the need for intellectual property rights such as patents. Support for entrepreneurship is provided mainly

to undergraduate students through the innovation center. Some ideas that help solve local problems in Azerbaijan have received project-based funding from oil-related companies like BP in the UK.

(ADA University) Through the KOICA ODA project, ADA University is operating a joint IT degree program with Soongsil University in Korea and promoting cooperation related to entrepreneurship. Although the university has an innovation center and is running projects to nurture and support start-ups, it lacks adequate funding to operate start-up programs, resulting in an education-centered start-up support system. ADA University aims to expand cooperation with Korea and Italy to develop start-up innovation capabilities. However, due to a lack of specialized entrepreneurship professors and difficulties in raising follow-up funds, practical incubation programs for start-ups have been temporarily suspended. Some telecommunications companies are interested in supporting start-ups, but no significant investments or partnerships have been established yet. If a fund support system that can support the actual creation of start-ups beyond entrepreneurship education is established, ADA University hopes to collaborate with any country or institution worldwide.

(Azerbaijan University of Architecture and Construction) The university participates in and advises on various government-led projects related to Sabah City, smart city construction, and other areas involving Azerbaijan. It seeks to establish exchanges with universities in Korea regarding earthquake and architectural design. The university operates six colleges and is interested in developing core competencies in innovation and entrepreneurship. They aim to apply the technology-based startup processes from Korea to their students at the innovation center and hope to involve Korean experts in their ongoing courses. Through HTP, they wish to participate in or develop future cooperation plans with Korea's K-Innovation projects.

(University of Economy) The University of Economy operated a bootcamp-style program in 2022 to nurture start-ups, discovering about 25 teams. These teams were provided with follow-up education and support programs linked to international funds such as ABB (Azerbaijan Bank of Business), which is one of the largest banks in Azerbaijan, similar to

Korea's Industrial Bank. However, there are still limitations in supporting initial seed investments. ABB supports the largest start-up incubation center and fund in Azerbaijan and operates hackathon-style entrepreneurship education programs. Unlike other universities, the University of Economy has a technology commercialization center that can file and support patents, filing about 16 patents annually, with approximately 50% being registered. In terms of cooperation with Korea, the university exchanges degree programs with Kyung Hee University and Woosong University and has partnerships with universities in Hungary and Greece.

(Baku Higher Oil School) Established in 2017 to revitalize the oil industry, train professionals, and promote industry-academia cooperation, Baku Higher Oil School is specialized in petrochemical fields. It resembles Korea's specialized science and technology universities in structure. Excellent students from Azerbaijan enroll, and many find employment with large global oil companies. Various companies, including the EU, ABB, and SOCAR, provide funds for industry-academia cooperation. The university operates six departments in the fields of oil and chemical engineering. The university's president and key executives show great interest in Korean start-ups. They conduct entrepreneurship and start-up-related education centered on lean start-ups and have partnerships with public and private start-up-related organizations such as the Innovation and Digital Development Agency, enactus, CRDF global, Horizon Europe, Start-up wise guys, Teknofest, Tubitak, Viveka, and Erasmus+.

1.2 Background of the Azerbaijan I-Corps Program Proposal

The Science and Technology Policy Institute has supported the creation of an innovation ecosystem in Azerbaijan over the past two years through K-Innovation. This includes establishing government governance, promoting industry-academia cooperation and open innovation, creating intellectual property rights and systems, and fostering an innovation ecosystem through technology start-ups in non-oil sectors through policy research and education.

I-Corps program among local experts in Azerbaijan, a TTT (Train the Trainer) approach will be designed to apply localized training programs and foster trainers.

For this, it is necessary to identify and nurture local experts (professors, science and technology policy and support institutions) who can influence the fostering of technology start-ups and innovative entrepreneurs locally. This process was also similar in Korea for the establishment of the I-Corps program.

Additionally, the main task of the third year, the Azerbaijan I-Corps program, requires persuading government departments to secure budgets through policy research centered on a small number of experts. It is also necessary to identify and nurture groups that can design student team education programs in line with the Korean model.

Selected TTT program experts need to share Korea's experiences and cases of developing localized textbooks and curricula for training trainers. It is important to establish a system for selecting teams, build a selection and evaluation system for technology start-up teams, support MVP development to enhance customer discovery in the market, and track and manage these activities and outcomes annually to establish a sustainable follow-up management and support system.

For this, the third year will consider the following aspects in operating the program:

(Train The Trainer Program) The main target for TTT recruitment should be similar to Korea, with a 1:1 ratio of university professors to market experts. University faculty should create and operate a pool of professors capable of educating university (graduate) students. However, there is still a lack of information on who teaches (technology) entrepreneurship at universities and their academic backgrounds. More research is needed on who would be suitable as TTT targets in Azerbaijan among other expert groups.

(Selection of Program Trainee) It is necessary to establish a direction for who will be targeted for education. Initially, it is essential to encourage the participation of university professors and industry experts and select those who can contribute to the program's spread. For example, institutions where already established TTCP experts belong or government start-

up support agencies should be persuaded to support MVP costs with small funds, serving as a platform to discover and attract prospective technology entrepreneurs. While there may be some initial support from countries like Korea, it is essential for the self-sufficiency of the Azerbaijan I-Corps training program in the mid to long term.

(Curriculum and Duration) The 2024 I-Corps TTT training program curriculum is designed in three stages: ① a two-day online pre-education program for selecting participants, ② a two-day offline intensive bootcamp, ③ an online office hour program conducted three times over six weeks, and ④ a final presentation evaluation and certificate award. The total duration is approximately three months.

(Others) It is necessary to develop Azerbaijan standard textbooks and curricula through instructors and recycle successful cases for education (non-oil industry sectors). Networking cooperation with Korean start-up-related institutions (e.g., Korea Youth Entrepreneurship Foundation, KoEF) is needed to ensure continued exchange in the field of technology start-up education even after the Science and Technology Policy Institute's project en

2. Current Status of the I-corps Programs in the U.S. and Korea

2.1 Introduction to the U.S. NSF I-Corps Programs

The U.S. NSF (National Science Foundation) I-Corps™ program served as the model for the Korean I-Corps program. It was introduced to Korea in 2014 through negotiations led by Korea's Ministry of Science and ICT, making Korea the first foreign country to adopt the program.

NSF I-Corps aims to train academic researchers in entrepreneurial skills under the American Innovation and Competitiveness Act. For over 70 years, NSF has led scientific research, innovation, and education, contributing to scientific advancement, national health, prosperity, security, and enhancing America's competitiveness on the global stage. Basic scientific discoveries and applied research in science, engineering, and education have led NSF to develop various technologies that benefit society.

In March 2022, NSF established the Directorate for Technology, Innovation, and Partnerships (TIP) to promote technology transfer and innovation. TIP was created under the CHIPS and Science Act to generate new industries and high-wage jobs in STEM fields and solve industrial problems.

TIP's Lab-to-Market platform provides researchers with options to translate technology into products and services with social and economic impact. It includes programs like NSF I-Corps (Innovation Corps), Partnerships for Innovation (PFI), and the America Seed Fund.

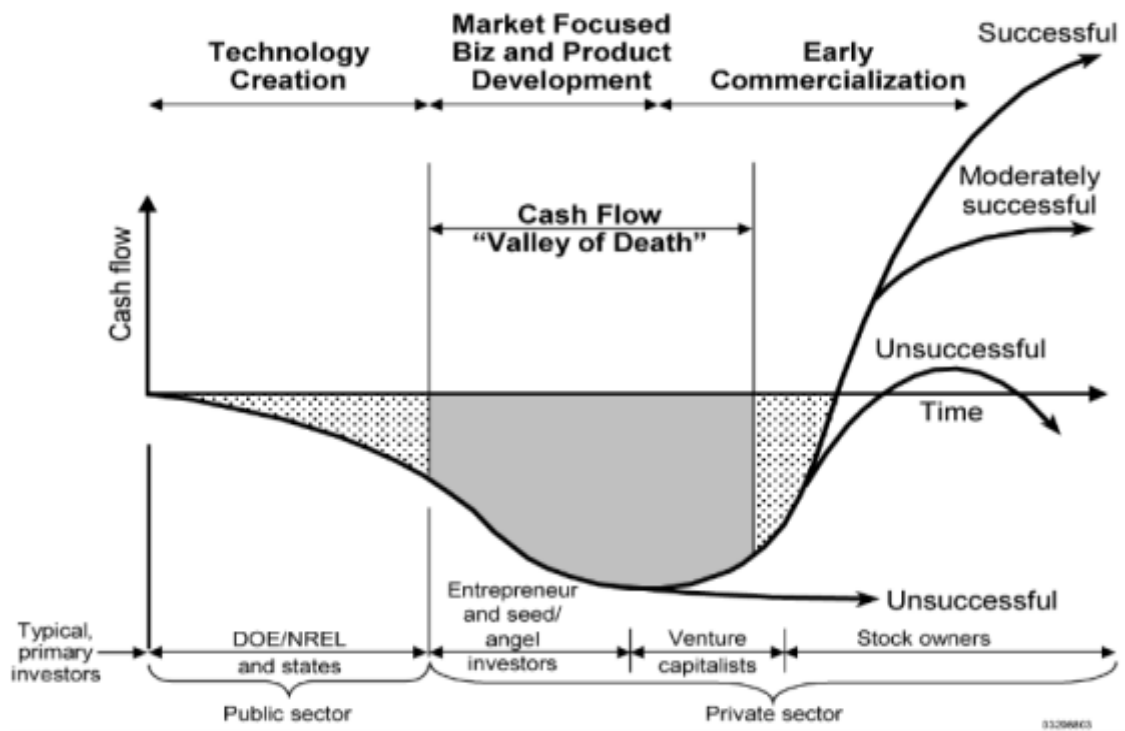
TIP also supports new translation pathways like the Pathways to Enable Open-Source Ecosystems (POSE) program, which accelerates the economic and social potential of projects born from basic science and engineering, creating benefits across the national innovation ecosystem.

TIP focuses on bringing together researchers, practitioners, and users to determine research directions, promote iterative co-design and co-creation, develop innovative technologies and solutions for national social and economic challenges, and foster future talent.

2.1.1 Background of the U.S. NSF I-Corps Program

Startups initiate the commercialization of new technologies, promote industrial development, and have a direct economic impact. NSF I-Corps teaches researchers how to lead discoveries from the lab to the market, encouraging them to recognize the unique requirements of technology translation and the characteristics and opportunities of startups and SMEs as part of the “Lean Startup” movement, which is the optimal method to overcome the “Valley of Death.”

[Figure 3-1] Startup Stage the Valley of Death



Source: L.Murphy-P.Edwards (2003)

Customer Discovery, Hypothesis Testing, Pivoting Based on New Knowledge PIVOT Pivot (PIVOT): In business, a “pivot” refers to strategically changing direction when a company is not meeting expectations, facing market changes, receiving customer feedback, or discovering new opportunities. This often involves redefining the company’s original value proposition, product offerings, target customers, or current business model. Pivoting is a

powerful strategy that helps startups or small businesses overcome stagnation, competitive pressure, or difficulties in scaling, allowing them to better meet market demands and maintain growth potential.

2.1.2 NSF I-Corps Program's Main Directions

The U.S. National Science Foundation (NSF) advances the economic and social potential of basic science and engineering research and develops the national innovation ecosystem through the NSF I-Corps (Innovation Corps) program.

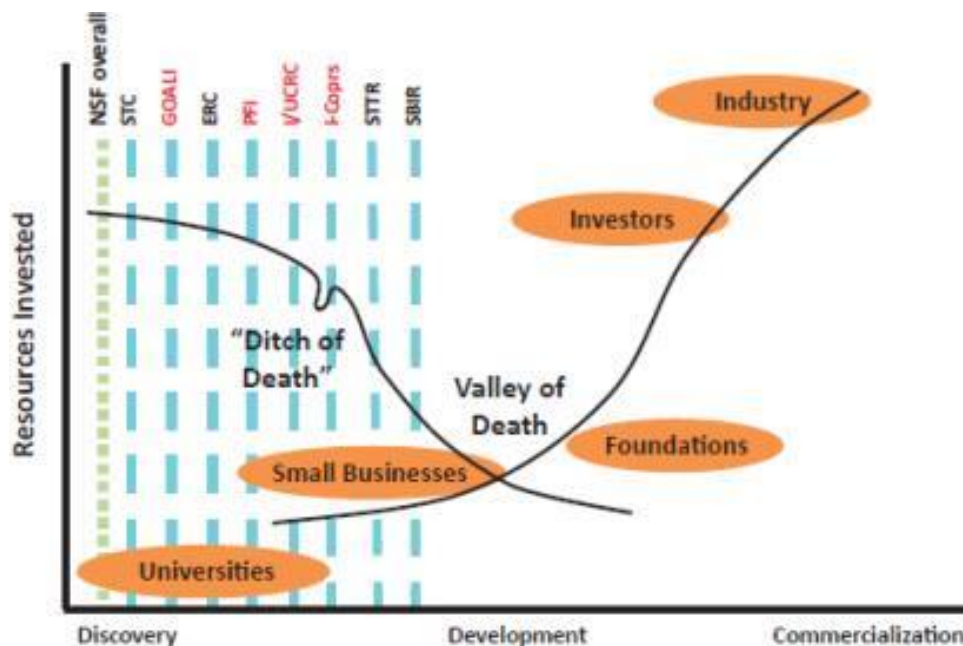
The core program is the NSF National I-Corps program, which aims to realize the economic and social potential of basic science and engineering research projects and accelerate the creation of various levels of benefits across the national innovation ecosystem. It aims to train personnel equipped with the technical expertise and industry analysis and prediction capabilities needed by U.S. research companies.

Additionally, the National Institutes of Health (NIH) and the Department of Energy (DOE) also implement I-Corps programs tailored to their respective fields, focusing on the following four areas to promote performance:

- **Training an Entrepreneurial Workforce:** Training researchers with capabilities in customer discovery, hypothesis testing, and minimum viable product development to increase the likelihood that basic scientific research will lead to economic and social impact
- **Translating Technologies:** Applying researchers' scientific discoveries to the market by turning them into products or services
- **Enabling Economic Impact:** Creating economic impact through the successful establishment of startups and subsequent funding support
- **Nurturing an Innovation Ecosystem:** Emphasizing diversity and inclusion while building various infrastructures within the innovation ecosystem

NSF I-Corps aims for inclusive economic prosperity through its programs and currently offers educational training courses to connect the achievements created through two programs to Lab to Market. The first program targets academic researchers through the National I-Corps Teams program, and the second targets companies at the NSF SBIR/STTR stage through the Small Business Innovation Research (SBIR) and Small Business Technology Transfer (STTR) I-Corps programs.

[Figure 3-2] NSF's Innovation Bridge Structure



Source: NSF(2024)

The I-Corps program aims to transform inventions into societal impacts through a seven-week experiential training course led by technology transfer experts from the National Science Foundation (NSF).

The standardized curriculum integrates scientific inquiry and customer discovery, focusing on building a data-driven culture characterized by rigor, relevance, evidence, and inclusivity.

By participating in I-Corps training, researchers are expected to shorten the time it takes to transition promising ideas from the lab to the market. Through this process, NSF aims to foster innovation among science faculty and students, encourage regional coordination,

networking, and sharing of promising practices, and strengthen the national innovation ecosystem to address societal needs and economic opportunities.

The NSF I-Corps curriculum utilizes accelerated training courses applied from the Lean Startup methodology, focusing on identifying viable business models that consider both future research and commercialization opportunities.

Participants aim to explore application areas that can be addressed through technological innovation by conducting interviews with potential customers, partners, and other business stakeholders through team collaboration.

Teams conduct NSF I-Corps training with their peers, allowing them to learn from each other, develop interpersonal skills, and motivate additional business opportunities and success.

NSF released the biennial report ‘The NSF Innovation Corps (I-Corps) Biennial Report 2023’ in June 2023, summarizing the achievements of the NSF Innovation Corps (I-Corps) program up to 2022.

2.2 Korea’s I-Corps Program

2.2.1 Introduction to the Korean I-Corps Program

The Korean I-Corps program is modelled after the National Science Foundation (NSF) I-Corps program in the United States. It aims to provide market exploration education specialized in science and technology entrepreneurship to pre-startup teams based on university or research achievements. From 2015 to 2016, a pilot project was conducted to foster technology start up teams centered around the Korea Advanced Institute of Science and Technology (KAIST) and Pohang University of Science and Technology (POSTECH).

Starting in 2017, the program expanded to general universities through the “Entrepreneurship-Centered University” project. Since 2018, it has spread nationwide under the name “Public Technology-Based Market-Linked Startup Exploration Support Project,” with major hub universities (Node universities) across the country.

This program supports pre-startup teams in conducting interviews to verify the feasibility of their business models with domestic and international potential customers, and helps them modify and improve their startup items based on the feedback.

Additionally, the Korean I-Corps program offers various accelerator programs and technology commercialization consulting services to support domestic startups. It also provides tailored growth management programs to help Korean startups enter the U.S. market.

As of 2024, there are seven hub universities (Node universities) supporting this program nationwide, detailed as follows:

- Supervising Ministry/Executing Agency: Ministry of Science and ICT/ Commercialization Promotion Agency for R&D Outcomes (COMPA)
- Basic/Practical Program Operating Agency in Korea: Korea Entrepreneurship Foundation (KoEF)
- Overseas Practical Program Operating Agency: KIC D.C (based in Washington, D.C.)
- Seoul Metropolitan Area: Korea University (covering northern Seoul and Gangwon Province), Sungkyunkwan University (covering southern Seoul, Incheon, and Gyeonggi Province)
- Chungcheong Area: Korea Advanced Institute of Science and Technology (KAIST)
- Honam Area: Gwangju Institute of Science and Technology (DGIST)
- Daegu and Gyeongbuk Area: Ulsan National Institute of Science and Technology (UNIST), Pohang University of Science and Technology (POSTECH)
- Women-Specialized Node: Ewha Womans University

2.2.2 Performance of the Korean I-Corps Program

From 2015 to 2021, approximately 42.7% (186 out of 435) of the teams that completed the training established startup companies by 2022, approaching the approximately 50% level of the U.S. NSF I-Corps.

About 65.5% (122) of the startups are still surviving, showing a three-year survival rate of about 70% and a five-year survival rate of about 44.5%. In terms of job creation, 1,003 new jobs were created, with an average of 5.4 employees hired per startup.

For a direct comparison with the U.S. I-Corps training program, the performance from 2017 to 2020 was divided into two-year intervals. Between 2017 and 2018, the amount of private investment raised was approximately 157.77 billion KRW, accounting for about 97.8% of the total, with government support funds centered around TIPS accounting for about 2.2%. Between 2019 and 2020, the amount of private investment slightly decreased to about 58.54 billion KRW, accounting for about 92.1% of the total investment raised.

As of August 2023, the total investment raised by surviving companies (2017-2020) was approximately 224.8 billion KRW through 24 companies, which is about 9.56 times the government support budget of 23.5 billion KRW, indicating high efficiency.

This is an excellent performance, comparable to the total investment raised by U.S. NSF I-Corps startups during the same period, which was 262 million USD (approximately 340.6 billion KRW), when considering the budget and the number of supported teams.

The program aims to quickly deliver science and technology to areas where it is needed by the public.

2.2.3 Progress of the Korean I-Corps TTT Program

From 2015 to July 2024, 740 teams have completed the Korean I-Corps program, resulting in the creation of outstanding science and technology startups such as Luxrobo (Pre-IPO), Medino (Series B), and Classting (Series B).

For instructors (Korean I-Corps Instructors), an average of about 8 I-Corps Instructors have been trained annually through an agreement with George Washington University in the U.S. since 2017, with a total of around 46 instructors capable of guiding Korean technology startup teams as of 2022.

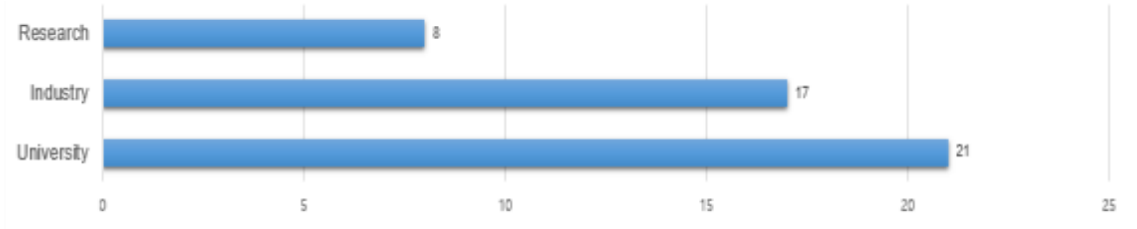
[Table 3-1] Instructors in Korea

Year	2016	2017	2018	2019	2020	2021	2022	Total
Instructors	7	11	8	7	7	COVID-19	6	46

Source: KoEF(2024)

For Korean instructors, to expand the education of technology entrepreneurship at universities, approximately 45.6% of the 46 instructors were selected as university professors, 36.9% were experts from the industry, and the remaining 17.3% were instructors who excelled in fields such as research.

[Figure 3-3] Korean Instructions by Jobs



Source: KoEF(2024)

2.2.4 Implementation System of the Korean I-Corps TTT

2.2.4.1 Government Implementation System for the I-Corps Project for Science and Technology Startups

In the case of Korea's TTT program, the Korea Commercialization Promotion Agency (COMPA) and the Korea Youth Entrepreneurship Foundation (KOEf), which are the actual implementing agencies of the project, recruit a certain number of participants annually. Until 2022, they received instructor training at George Washington University in the United States (George Washington University is one of the NSF I-Corps National Node universities).

After completing the TTT training in the United States, the Korea Youth Entrepreneurship Foundation supports instructors and assigns them to educational training programs linked to domestic I-Corps or provides necessary support.

In Korea, from 2017 to 2022, around 8 Korean instructors were selected annually and supported to undergo the TTT (Train the Trainer) training course at George Washington University, one of the NSF Node universities in the United States.

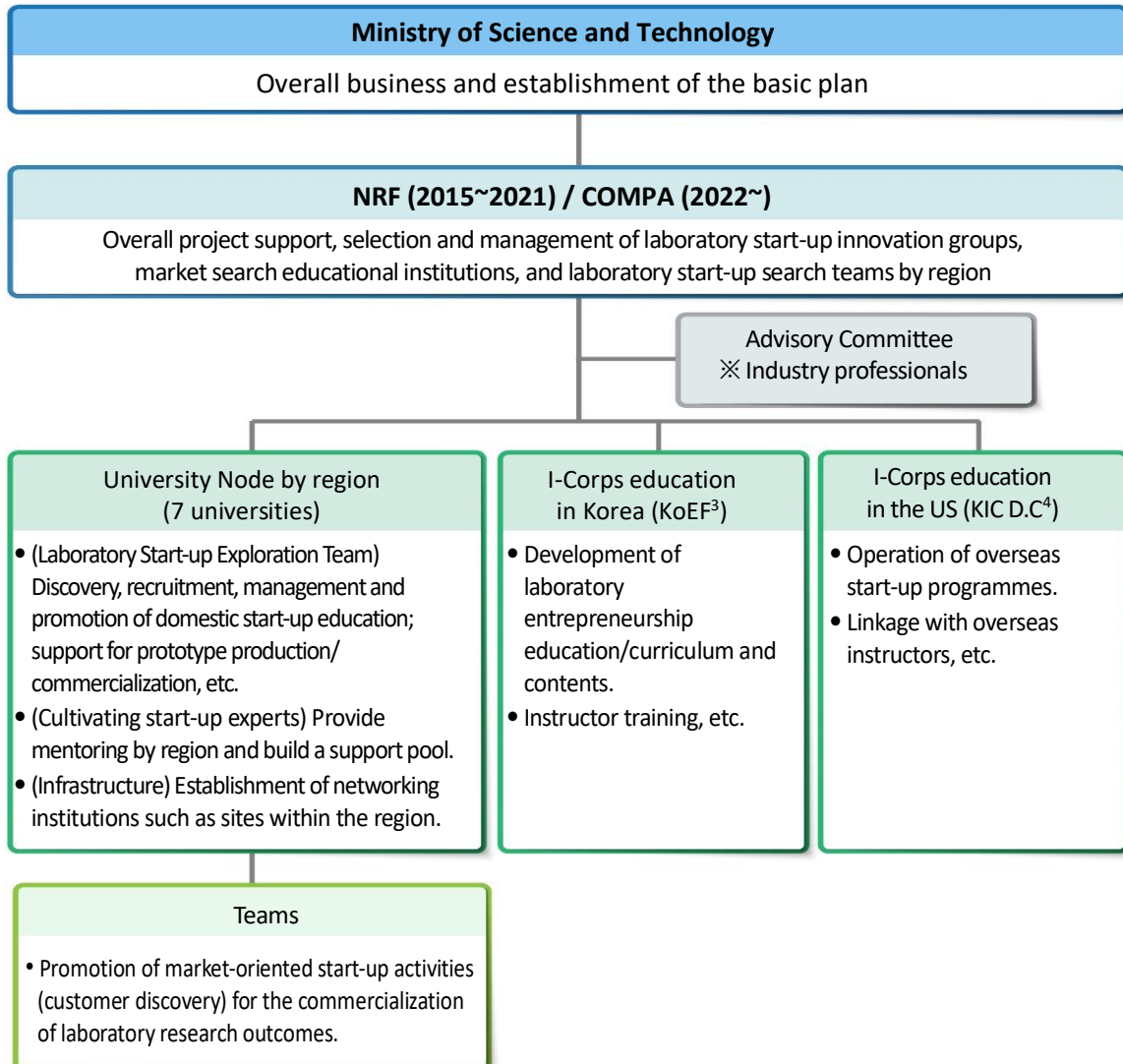
By the end of 2022, approximately 46 instructors had completed the TTT training course. Instructors are primarily selected from ① university professors, ② startup professional instructors, ③ professionals such as patent attorneys, lawyers, and accountants, and ④ highly experienced individuals with business or industry experience.

Considering the impact of COVID-19 and the sufficient number of trained Korean instructors, the selection target for instructors has been shifted from university professors to industry-specific mentors since 2023.

Additionally, since 2023, the instructor grades have been divided into three levels: Master Instructor, Instructor, and Pre-Instructor. This system operates similarly to a promotion system, where industry-specific mentors are granted Pre-Instructor qualifications and can be adjusted to higher levels based on participation in I-Corps and other startup education programs after completing the TTT certification training course.

Korean instructors trained through the TTT program are active as instructors centered around Node universities and participate in activities to recruit laboratory startup exploration teams through each Node university's Mini I-Corps program before the official I-Corps program is launched in Korea.

[Figure 3-4] Structure of the Korea I-Corps programme, NRF



Source: NRF & GS SHIM (2022)⁵

³ KoEF (Korea Entrepreneurship Foundation): <http://www.koef.or.kr/> (Date accessed: 10 October 2024)

⁴ KIC D.C (Korea Innovation Centre Washington D.C): <https://www.facebook.com/KoreaInnovationCentreDC/> (Date accessed: 10 October 2024)

⁵ NRF & GS SHIM (2022), "Policy research of the Korean I-Corps Programme," NRF, January 2022., p. 103.

2.2.4.2 Structure for Supporting Instructors with in the Implementation System

When detailing the implementation system, the structure for fostering and supporting instructors is organized under the following roles and responsibilities.

[Table 3-2] Korea I-Corps Program Stackholders

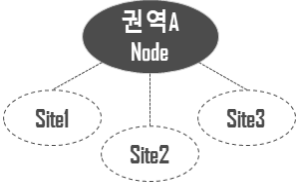
	Group	Role	Roles and Responsibilities
Stakeholders	Node Universities	Start-up Incubation and support	• Discover and nurture a start-up search team (intensive education, pivoting, etc.). • Establish a mentor pool for each region and support mentoring. • Establish a site within the region and discover overseas market discovery educational institutions.
	KoEF	Education (Local)	• Develop a domestic entrepreneurship education curriculum, domestic common start-up education, and basic and practical start-up education for domestic start-up search teams. • Discover, nurture and support instructors.
	KIC D.C	Education (Global)	• Conduct basic and practical start-up education for overseas start-up search teams. • After overseas training, support additional mentoring and incubation by overseas start-up experts. • Linkage with international instructors.
Professionals	Mentors	Scale-up (Pivoting)	• Pivot and supplement the directions for start-up business items. • Assist with discovering customers and share start-up know-how.
	Instructors	I-Corps methodology training	• Customer Discovery Methodology Training • Train the start-up search team (enhance their entrepreneurial mindset, etc.)

Source: NRF & GS SHIM(2022)⁶

⁶ NRF & GS SHIM (2022), "Policy research of the Korean I-Corps Programme", NRF, January 2022., pp.107.

Additionally, the hub universities, referred to as Node universities, are structured to discover and nurture technology startup teams from other universities within their region or to conduct educational and team discovery activities for prospective technology startup teams through the operation of the Mini I-Corps program.

[Table 3-3] Role of Node and Site University

<Structure >		(Node) University	(Site) University
	Condition	Regional university as a business incubation agency.	Belongs to the Node University of the region.
	Major role	- Recruitment of technology-based start-ups for the region. - Start-up incubation programme (around 5 months).	- Recruitment of technology-based start-ups and introduction of the teams to the Node University. - Mini I-Corps programme (2-5 weeks).
	Follow-up	Follow-up for excellent teams with other start-up programmes.	Follow-up for excellent teams that enter the I-Corps programme.

Source: NRF(2022)⁷

⁷ NRF & GS SHIM (2022), "Policy research of the Korean I-Corps Programme", NRF, January 2022., pp.108.



CHAPTER 4

Strategic Plans of the Azerbaijan Model I-Corps TTT Program

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Chapter 4. Strategic Plans of the Azerbaijan Model I-Corps TTT Program

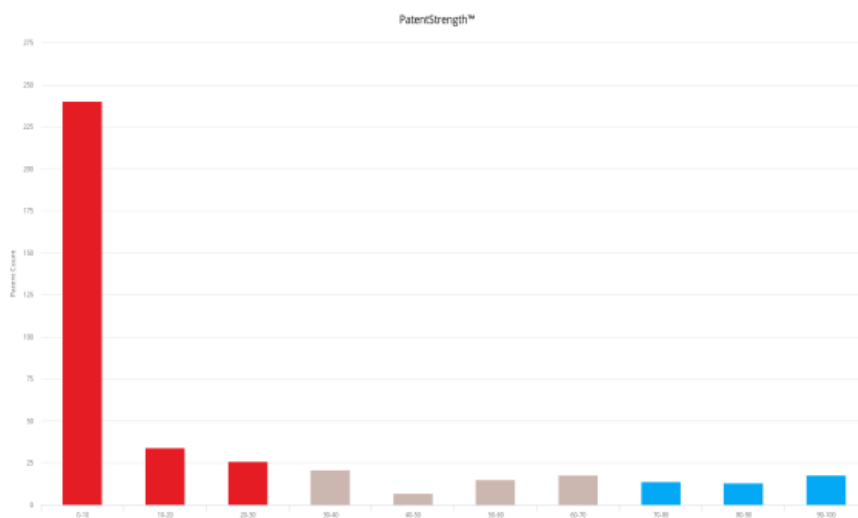
1. Strategic Plan of the Azerbaijan Model of the Azerbaijan Model I-Corps TTT Program

1.1 Direction for Planning the TTT(Train the Trainer) Program to Foster the Technology Start-up Ecosystem in Azerbaijan

1.1.1 Selection of Technologies for the Azerbaijan I-Corps TTT Program

In the case of Azerbaijan, to localize and apply the I-Corps program, TTT (Train the Trainer) training will be conducted focusing on the personnel from HTP and related institutions participating in this project. Using big data systems, a framework will be established to nurture startup pre-entrepreneurs by utilizing innovative patents registered in non-oil sectors in Azerbaijan.

[Figure 4-1] Analysis of Patent Technology Intensity in Azerbaijan

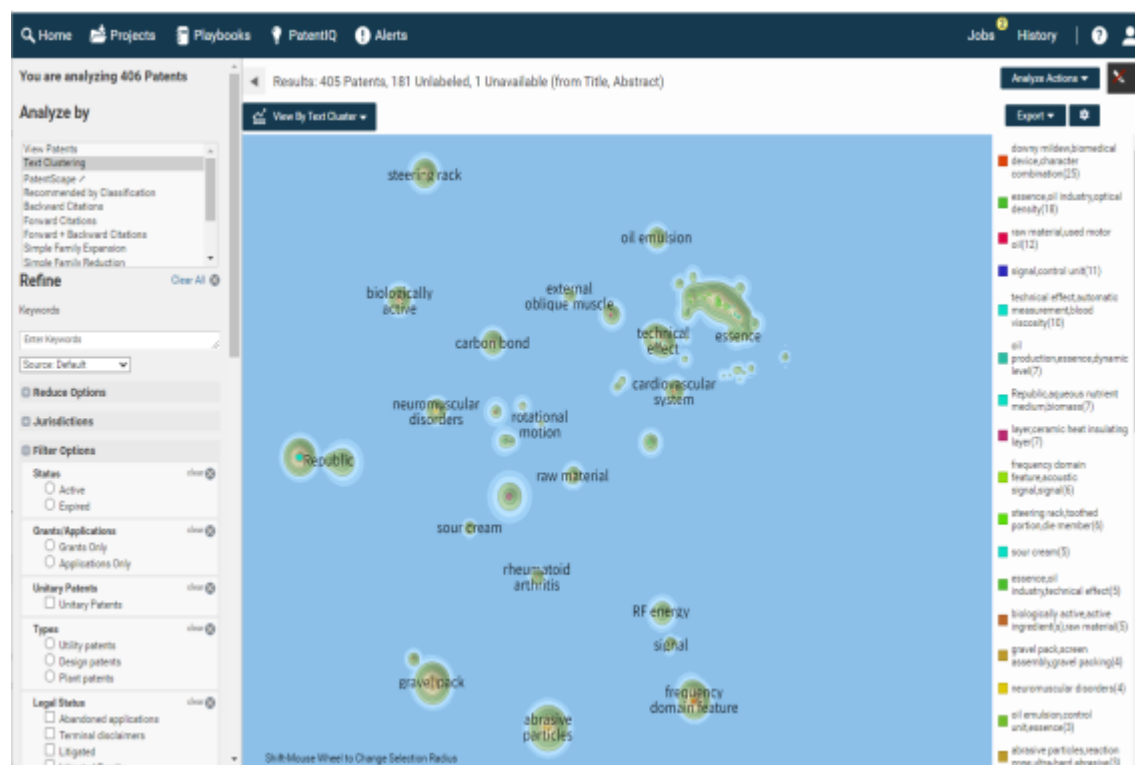


Source: GS SHIM (2024)

Through big data, the cumulative number of patents registered locally in Azerbaijan is approximately 405. Refer to the appendix for the complete list of target technologies. In this I-Corps TTT course, patents developed based on laboratories will be selected and linked to the I-Corps training program, designing the training program accordingly.

Additionally, in the process of selecting patents for technology startups, six target technologies will be selected based on practicality rather than focusing on high technology intensity patents, to induce technology startups in non-petrochemical fields and to help Azerbaijani instructors understand the training process.

[Figure 4-2] IP Portfolio in Azerbaijan



Source: GS SHIM (2024)

The selected six technologies will be autonomously chosen by the teams after the selection of participants, and basic patent information and technical content will be shared.

(A) Criteria for Target Patents of Technological Startups

For the I-Corps training program, target technologies were selected from the above patents including issued patents, pending applications, and abandoned applications. In particular, for the purposes of enhancing the curiosities of the Azerbaijan trainers, the patents were to satisfy one of the following requirements:

- At least one of the listed inventors is an Azerbaijan citizen;
- At least one of the listed assignees is an Azerbaijan citizen or organization;
- The patent is filed in the Azerbaijan Patent Office; regardless of the number of Azerbaijan inventors or Azerbaijan assignees;

In addition, the patents were also to satisfy one of the following additional requirements:

- Patents/applications issued or filed in the Azerbaijan Patent Office;
- Patents/applications issued or filed in the Eurasian Patent Office;
- PCT applications
- Other patents/applications if at least one of the inventors or assignees satisfies the above requirements

With the above requirements, “Innography IP Intelligence Software” of Clarivate were used to select the target patents, and the results included a total of 424 patents, issued, filed, and abandoned. Following Figure 3-00 shows an example of the search result produced by the Innography Software.

[Figure 4-3] An Exemplary Search Result of Innography IP Intelligence Software



Source: YT SHIM (2024)

The Innography IP Intelligence Software also provided strengths of the total of 424 patents (including issued, filed, and abandoned), where Clarivate explained that the “strength” of each patent is determined based on about 30 indices, the details of which Clarivate did not disclose. The strengths of 424 patents provided by the Innography Software are:

- Strength 90~100: 22 patents;
- Strength 80~90: 7 patents;
- Strength 70~80: 16 patents;
- Strength 1~70: 379 patents;

Once a pool of target patents was formed, additional considerations were established for training the Azerbaijan trainers, for the essence of the training is based on the Customer Discovery methodology. In particular, the trainees were to conduct in-depth interviews with the real customers. Accordingly, the patents which may require advanced knowledge or understanding of a particular technology sector were excluded in selecting the SIX target

patents. In contrary, the following technology sectors were given preference in selecting the SIX target patents:

- The patents relating to minerals or foods which are unique to Azerbaijan, for these sectors would enhance the interest of the Azerbaijan trainees;
- The patents relating to those technologies invented or developed by Azerbaijan universities or research institutes, for the trainees would find it more convenient to perform in-depth interviews with the real inventors or researchers;

[Table 4-1] Target Patents after 1st Screening

Patent intensity	Technology sector	Azerbaijan inventors or assignees	Patent Title (simplified)
1~70	Mechanical	YES	All-wheel-drive bicycle
1~70	Bio	YES	Method for producing emulsion cosmetic concentrate of water-oil type from sea buckthorn fruits
1~70	Bio	NO	Asteromonas salina mirzaladi strain for production of biomass
1~70	EE	NO	MCP telephone internet surfing and listening system and method, and call center
1~70	Software	NO	Auxiliary communication device and auxiliary communication method
1~70	Environmental/ Chemical	YES	Method for preparing a fertilizer from household waste
1~70	Bio	YES	Thermophilic strain of blue-green algae spirulina platensis Baku for production of biomass
1~70	Software	NO	Distributed computer system for document authentication
1~70	Mechanical	NO	Peeling device and method for peeling foodstuffs
1~70	Mechanical	NO	A keypad array of mobile phone for inputting characters of Russian language
1~70	Aviation	YES	Method for disinfecting aircraft cabins and device for carrying out said method

Patent intensity	Technology sector	Azerbaijan inventors or assignees	Patent Title (simplified)
1~70	Game	NO	Word-map of match block
70~80	Business Method	NO	Authorization of cardless payment transactions
70~80	Communication	NO	Discovery and communication using direct radio signal communication
80~90	Mechanical	NO	Electronic device holder

Source: YT SHIM (2024) (based on the analysis by Innography IP Intelligence Software)

With the above potential target patents, the meetings were held to qualitatively analyze whether each patent of the above table would be suitable in training the Azerbaijan trainers, for the training period was relatively shorter than the regular training period. As a result, the following criteria were also adopted in finalizing the SIX target patents:

- Exclude patents related to heavy device industry;
- Exclude patents in complicated mechanical sector such as engines, transmissions, automation equipment, etc.;
- Exclude patents related to various parts of the engines, transmissions, automation equipment, etc.;
- Exclude patents related to sectors requiring advanced technological expertises such as automobiles, airplanes, petroleum industry, etc.;
- Exclude patents related to semiconductors;
- Exclude patents related to optics;
- Exclude patents related to advanced electrical engineering;
- Exclude patents related to certain chemistry, materials, etc.;
- Exclude patents related to pharmacologic sectors, particularly those without any human clinical trial data;

- Exclude patents related to medical imaging;
- Exclude patents related to medicine, particularly methods of performing surgery;
- Exclude patents related to sectors in which objective or quantitative determinations are not feasible, where such sectors include taste of food, texture of food, etc.;
- Exclude patents related to food processing; and
- Exclude patents related to sectors which are no longer technologically viable.

(B) Six Target Patents for the Trainees of the Technological Startups

Through discussions, we selected the following SIX patents as the target patents suitable for the Azerbaijan trainees. Details of the technologies related to such patents, main features in selecting such patents, and pros/cons of each patent are provided in detail below.

(B-1) Project 1 and Related Patent for Trainees of I-Corps TTT

Title: Eco-friendly organic fertilizers made from municipal garbages and oil/gas waste water

Technology Sector: environment, waste materials, waste water treatment

Features:

1. The patent was filed by the Institute of Azerbaijan National Academy to the Eurasian Patent Convention.
2. All inventors are Azerbaijan citizens and, therefore, it would be extremely handy for the trainees to perform in-depth interviews with the real inventors or researchers.
3. The patented technology relates to the method of manufacturing the organic fertilizers using the waste water from the oil and gas manufacturing facilities which is existing in Azerbaijan. Thus, this project would be able to attract the interest of the Azerbaijan trainees.
4. The downside of this patent, however, is that the patent does not provide detailed or in-depth information as to the types of municipal garbages suitable for manufacturing

such fertilizers, the technology of classifying numerous types of garbages and providing the raw materials for the fertilizers, the technology for transporting the oil/gas waste water to the site of manufacturing the fertilizers, and so on.

5. As a result, it may be difficult for the trainees or the potential technological startup founders to determine practicality or viability of the technology.

Related Patents:

1. EA 201700260A which was filed to the Eurasian Patent Convention and which is now abandoned. As a result, the patent does not possess any exclusive rights.

[Figure 4-4] Description of Project 1 as provided to the trainees

PROJECT 01 ECO-FRIENDLY ORGANIC FERTILIZERS USING MUNICIPAL AND OIL/GAS INDUSTRIAL WASTES Project Summary This project relates to equipment and methods of manufacturing fertilizers from municipal wastes and wastes from the oil/gas industry. More particularly, the organic components of the municipal waste are separated and then mixed with oil/gas drilling water which emerges from the oil/gas wells. As a result, this technology not only solves many problems associated with municipal waste managements but also provides fertilizers which is rich in minerals such as, e.g., boron. Background and Commercial Needs Household waste is evolving into a core urban challenge. In almost every country, increasing population generates more wastes, and it is not surprising that the quantity and volume of such wastes are ever increasing. In addition, as the food manufacturers employ fancy packages for delicate food, such waste also tends to contain toxic and hazardous elements.  Landfill Sites of municipal waste Many processes have been proposed to recycle the municipal wastes and to convert them into fertilizers. However, conventional processes suffer from various technical, commercial or hygienic deficiencies. Presently, household waste are transported to landfill sites and dumped into excavated garbage valleys.  Offshore drilling and pollution Oil/gas industry produces oil/gas drilling water during oil/gas exploration or production. When oil/gas is produced off-shore, oil/gas drilling water poses great environmental challenges. Discharges of water-based and low-toxicity oil-based drilling muds can lead to ecological impacts at the population and community levels on the seafloor. Accordingly, an improved method is need for eco-friendly disposing the municipal wastes as well as the oil/gas drilling water. Technical Details The advantage of this technology lies in the fact that oil/gas drilling water is used for the neutralization of household urban waste (such drilling water leaves the bowels with a temperature of 50~80°C), and	that shell rock is used as an alkaline reagent (shell rock results from accumulation of skeletal remains of organisms, shells, in the coastal parts of the seas). In particular, oil/gas drilling water in a temperature range of 50-80°C provide sufficient conditions for killing or compromising mesophiles and thermophiles which tend to reside in the organic components of the municipal waste. This technology aims to provide a simple process of obtaining fertilizer using affordable cheap reagents (mainly waste and natural substances) as raw materials, and to improve the quality of such fertilizer. An exemplary method of providing such fertilizer includes the steps of: 01. Separate inorganic inclusions from municipal waste; 02. Obtain organic material-rich municipal waste; 03. Mix this municipal waste with oil/gas drilling waters emerging from bowels in a temperature range of 50-80°C; (a) Typical oil/gas drilling water has the following composition (grams / liter): Br = 1.20~1.60; Cu = 0.5~1.3; Mg = 0.1~0.9; Sulfates = 0.1~1.3; Di-borate (B₂O₃) = 0.09~0.8; and Rest (water) = 95.1~98.0. (b) The organic component of the municipal waste and the oil/gas drilling water are taken in a ratio of about 1: 2.5~3.5. (c) Shell rock is added in an amount providing the resulting fertilizer with a pH of 6.5~7.5. 04. Obtain pulp by keeping the mixture at a temperature of 100~105°C for 1.5 hours; 05. Cool the pulp for 30~40 minutes; and 06. Neutralize the resulting pulp with an alkaline shell reagent. The proposed method provides the high-quality fertilizer containing such an important element as boron. To implement the method, simple equipment is used. The energy consumption for creating the thermal regime necessary to neutralize the bacteriological inclusions contained in municipal waste from the heat of the used oil drilling water is eliminated, and available shell rock, which is local raw material, is used. Further technical details are provided in E.A. Patent Serial No. 2017/00260A which is entitled "Method for preparing a fertilizer from household waste" (see below). Technical/Commercial Advantages or Challenges Through utilizing the organic components of the municipal waste and the oil/gas drilling water emanating from the oil/gas sites, this technology solves two headaches with one stone. The proposed technology seems to be particularly suited for a country which produces oil or gas such as, e.g., Azerbaijan. The patent covering this technology was filed on Feb. 23, 2017 but seems to have been abandoned thereafter. Therefore, this technology can be practiced without having to obtain a license from anybody. References 01. Method for preparing a fertilizer from household waste, EA201700260A 01'. ОПИСАНИЕ ИЗОБРЕТЕНИЯ К ЕВРАЗИЙСКОМУ ПАТЕНТУ, Евразийское патентное ведомство 032766 B1
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Source: YT SHIM (2024)

(B-2) Project 2 and Related Patent for Trainees of I-Corps TTT

Title: Method of performing credit card transaction without having to present the credit card to a merchant

Technology Sector: business method, computer networks

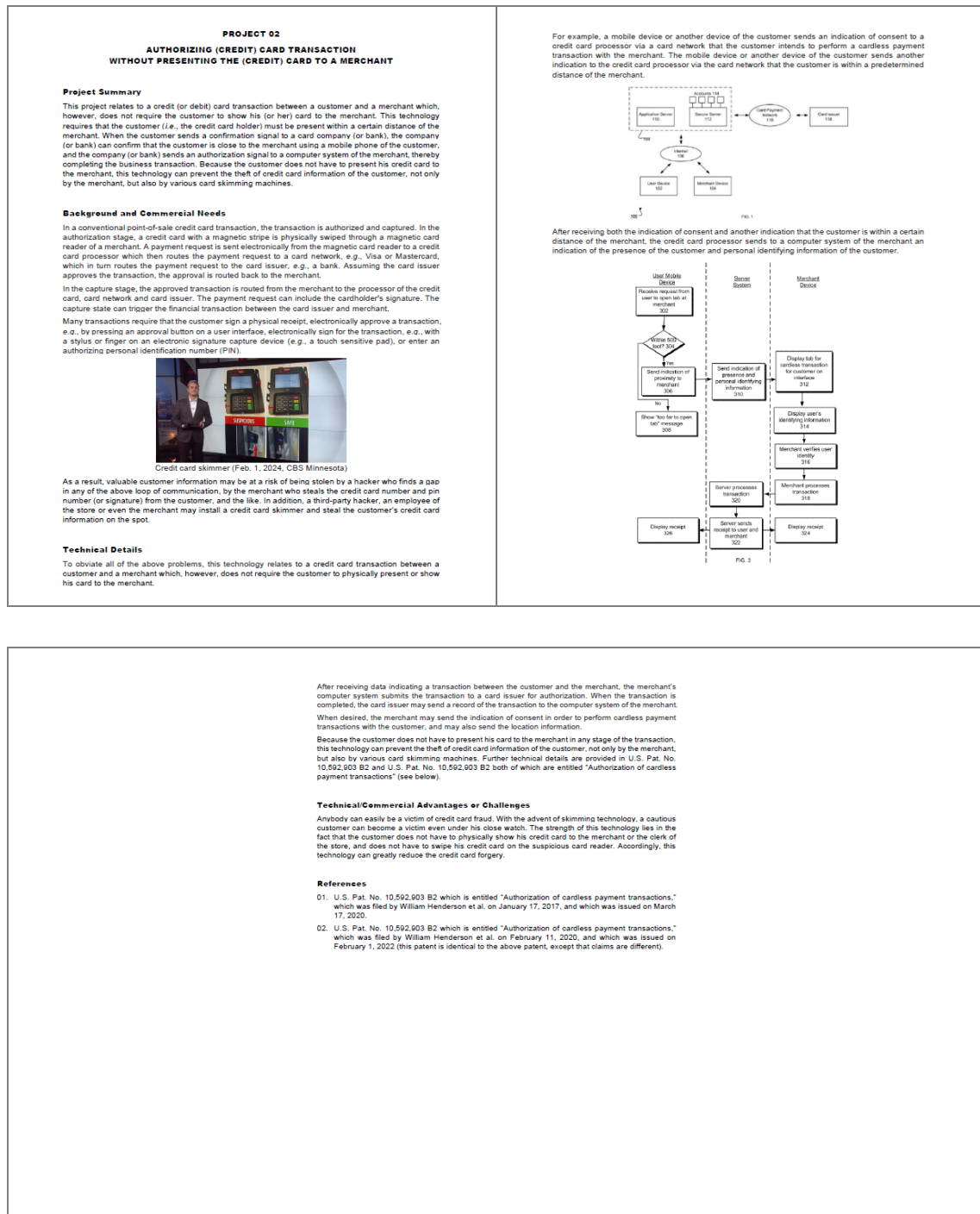
Features:

1. The patent was filed by the U.S. inventors to the U.S. Patent and Trademark Office.
2. The patent relates to a business method in which an owner of a credit card can perform a credit card transaction without having to present the credit card to a merchant. As a result, the patent claims that the security and privacy of the credit card holder can be maintained secret to the merchant.
3. To this end, the patent requires a unique network which can identify where the credit card holder is on real time (or at the instance of business transaction) such that the card company (or bank) can determine whether or not the transaction is the one requested by the legitimate card holder.
4. The related technology requires basic knowledge in computer network and business transaction.
5. Since the patent does not require any specific or advanced knowledge of computer network, it may be convenient to find qualified customers to perform the in-depth interview. Accordingly, the Azerbaijan trainees who do not have concrete background in computer science can take this project.

Related Patents:

1. U.S. Pat. No. 10,592,903 (issued and still in force).

[Figure 4-5] Description of Project 2 as provided to the trainees



Source: YT SHIM (2024)

(B-3) Project 3 and Related Patent for Trainees of I-Corps TTT

Title: Method of authenticating a document using distributive network

Technology Sector: business method, software

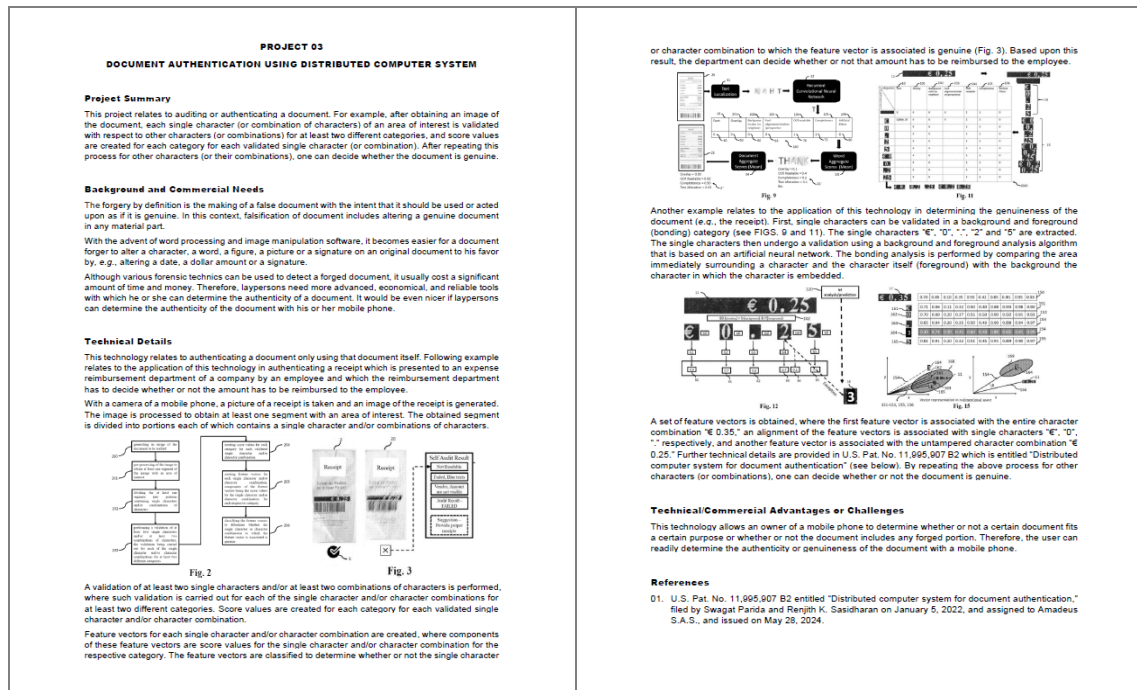
Features:

1. The patent was filed by the French inventors to the U.S. Patent and Trademark Office.
2. The patented technology includes the steps of: (1) establishing authentication criteria; (2) obtaining an image of a document to be authenticated; (3) comparing multiple pixels or characters as obtained in the images according to the established criteria; and (4) obtaining a score from such comparisons; and (5) determining whether or not the document or a certain portion of the document is genuine or falsified.
3. The related technology requires basic knowledge in business method and software in general.
4. Since the patent does not require any specific or advanced knowledge of software, it may be convenient to find qualified customers to perform the in-depth interview. Accordingly, the Azerbaijan trainees who do not have concrete background in software can take this project.

Related Patents:

1. U.S. Pat. No. 11,995,907 (issued and still in force).

[Figure 4-6] Description of Project 3 as provided to the trainees



Source: YT SHIM (2024)

(B-4) Project 4 and Related Patent for Trainees of I-Corps TTT

Title: A keypad of a mobile phone specifically adapted to inputting the Russian alphabets

Technology Sector: mobile phone, user interface

Features:

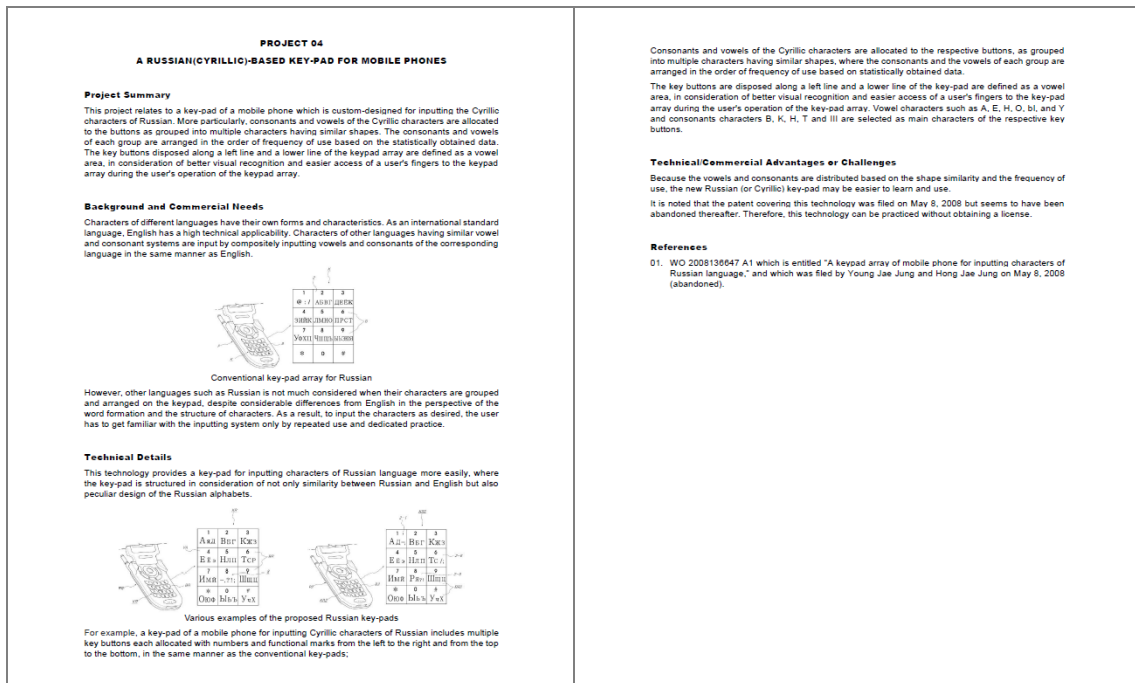
1. The patent was filed by a Korean engineer of LG EElectronics to the Korean Intellectual Property Office and to the PCT.
2. The patented technology claims that each Russian alphabet is assigned to each key of a QWERTY keypad according to the frequency of uses, shapes of the Russian alphabets, and so on, such that a user may readily input the Russian alphabets to the QWERTY pad.
3. The Azerbaijan alpbets are generally similar to the Russian alphabets and, accordingly, this patented technology may attract attention of the Azerbaijan trainees.

- However, it is not certain how well-versed the Korean inventor is in Russian, and the practicality of this technology may not be justifiable.

Related Patents:

- WO 2008136647 A1 (not filed in foreign countries)

[Figure 4-7] Description of Project 4 as provided to the trainees



Source: YT SHIM (2024)

(B-5) Project 5 and Related Patent for Trainees of I-Corps TTT

Title: An apparatus and method of disinfecting a cabin of an airplane using ozone gas

Technology Sector: aviation, cleaning technology

Features:

- The patent was filed by the Institute of Azerbaijan National Aviation Academy according to the PCT.

2. All inventors are Azerbaijan citizens and, therefore, it would be extremely handy for the trainees to perform in-depth interviews with the real inventors or researchers.
3. It is however noted that such a technology seemed to have been existing and that a few patents had been issued in the similar arts.
4. Accordingly, the trainees may have to analyze this patented technology and to compare it with the existing prior art.

Related Patents:

1. A PCT has been filed but not filed in individual countries (no exclusive rights)

[Figure 4-8] Description of Project 5 as provided to the trainees

PROJECT 05
METHODS AND DEVICES FOR DISINFECTING AIRCRAFT CABINS WITH OZONE

Project Summary

This project relates to methods of disinfecting an aircraft cabin and device for carrying out such methods. More particularly, the methods include the steps of manually or mechanically cleaning the cabins and another step of blowing an O₃-air mixture with a certain O₃ concentration therethrough by micro-ports for a certain duration. Thus, the cabin and its compartments can be readily and effectively disinfected.

Background and Commercial Needs

As was evidenced during the 2020 Coronavirus pandemic, airlines were looking for ways to promote a sterile and safe travel experience for their customers. Current disinfection methods involve using a large crew to deep-clean an aircraft at the end of the day. However, this approach requires a lot of manpower and a huge time commitment considering how many airplanes fly each day.



Even when the airlines deployed huge manpower, it was still difficult to deep-clean hard-to-reach places on the plane such as, e.g., the seat back pockets, and tray tables in the upright and locked position.

Technical Details

After manually or mechanically cleaning an aircraft cabin, an ozone-air mixture is blown into the cabin by micro-ports whose duration is equal to 3 seconds at an interval therebetween of 10 seconds with the ozone concentration ranging from 10 to 20 mg/sec.

The ozone-air mixture is produced by a device which includes a power charging unit connected to an input of a power accumulating and control unit, where the outputs of the power accumulating and control unit are connected to a network consisting of the primary coil of a transformer and a tyristor whose control electrode receives a control signal from the output of the power accumulating and control unit.

№ of the micro-ports	№ of the micro-ports	№ of the micro-ports	№ of the micro-ports	№ of the micro-ports	№ of the micro-ports	№ of the micro-ports	№ of the micro-ports	№ of the micro-ports	№ of the micro-ports
1	2	3	4	5	6	7	8	9	10
10	20	30	40	50	60	70	80	90	100
11	22	33	44	55	66	77	88	99	110
12	24	36	48	60	72	84	96	108	120
13	26	39	52	65	78	91	104	117	130
14	28	42	56	70	84	98	112	126	140
15	30	45	60	75	90	105	120	135	150
16	32	48	64	80	96	112	128	144	160
17	34	51	68	85	102	119	136	153	170
18	36	54	72	90	108	126	144	162	180
19	38	57	75	95	115	135	155	175	190
20	40	60	80	100	120	140	160	180	200

The device also includes an ozone generator which is connected in parallel to the secondary coil of the transformer and is provided with sets consisting of two low-voltage plates and one high-voltage plate

which is disposed therebetween and encompassed by barriers. Thus, it is possible to reduce energy losses, exclude a resonance overvoltage in the secondary transformer coil, and to increase ozone production, where the transformer is embodied in the form a line-output transformer whose number of primary turns is determined by a resonance frequency and a charging resistance is embodied in the form of the coil resistance of a fan. As a result, hospital grade sterilization was achieved in less than an hour and there was a 99.9999% reduction in virus due to ozone treatment.

Technical/Commercial Advantages or Challenges

It is noted that the patent covering this technology was filed on Dec. 29, 2005 but seems to have been abandoned thereafter. Therefore, this technology can be practiced without obtaining a license.

However, this technology had recently been practiced by others such as, e.g., 2B Technologies (see below). Accordingly, this project may require technical advancement with respect to the patent covering this technology (WO 2005/122696A2) as well as the technology practiced by 2B Technologies.

References

01. WO 2005/122696 A2 entitled "Method for disinfecting aircraft cabins and device for carrying out said method," filed by Arif Pashayev et al. on Dec. 29, 2005, and assigned to Azerbaijan National Aviation Academy of the State (abandoned).

02. 2B Technologies, "Using Ozone to Disinfect Aircraft Cabins" (2020?)
https://2btech.in/wp-content/uploads/docs/application_notes/Application_Note_108-L_AircraftDisinfection.pdf (last visited on Aug. 14, 20224)

Source: YT SHIM (2024)

(B-6) Project 6 and Related Patent for Trainees of I-Corps TTT

Title: A method of manufacturing concentrated emulsion for cosmetics from Azerbaijan sea Buckthorn fruits.

Technology Sector: cosmetics, chemical (extraction) process

Features:

1. The patent was filed by the Institute of Azerbaijan National Academy according to the PCT.
2. All inventors are Azerbaijan citizens and, therefore, it would be extremely handy for the trainees to perform in-depth interviews with the real inventors or researchers.
3. The patented technology is not directed to the manufacture of concentrated emulsion for cosmetics from Azerbaijan sea Buckthorn fruits, but rather to the new process of making the same.
4. The patent claims that the concentrated emulsion for cosmetics from Azerbaijan sea Buckthorn fruits can be made by the new method which requires a smaller number of process steps and, therefore, which is more economical.

Related Patents:

1. EA 03984 B1 which was filed to the Eurasian Patent Convention and issued.

[Figure 4-9] Description of Project 6 as provided to the trainees

PROJECT 06 METHODS FOR PRODUCING EMULSION COSMETIC CONCENTRATE OF WATER/OIL TYPE FROM SEA BUCKTHORN FRUITS Project Summary This project relates to methods of producing cosmetic cream concentrates based on natural, biologically active components contained in sea buckthorn fruits, specifically, for producing emulsion creams of oil/water types. More particularly, such methods can produce cream concentrates from sea buckthorn fruit with a smaller number of technological steps and shorter time spent on their performance, using the economically beneficial raw materials. Background and Commercial Needs Sea buckthorn oil is a red-orange oil derived from sea buckthorn plants. The most commonly used species for this purpose is <i>Hippophae rhamnoides</i>. Species belonging to this genus accumulate lipids in the mesocarp (fruit pulp), so the oil can be extracted from either the seeds or the pulp. The resulting oils (seed oil and pulp oil, also called fruit or berry oil) are used in dietary supplements, nutraceuticals, cosmetics, and skin care products.  Sea buckthorn oil is frequently used in cosmetic products, especially topical products marketed for aging skin and dry skin. Sea buckthorn oil is used in some shampoos and other hair care products. It has sometimes been used as a type of topical medication for skin disorders, although there is no high-quality research proving effectiveness or safety of any such uses.^[1] Taking sea buckthorn oil as a dietary supplement does not help with eczema. Technical Details The method of obtaining a cosmetic cream concentrate from sea buckthorn fruits is as follows. At the first stage, sea buckthorn fruits are cleaned, washed from impurities, passed through a juicer and juice is obtained. The resulting juice is filtered. To prepare emulsions from juice, 400 ml of juice is placed in a 2-liter beaker, 800 ml of refined olive oil is added to it and, stirring constantly, the temperature of the mass is brought to 80 °C. After that, 130-133 g of T-2 emulsifier is added to this mass and stirred until the emulsifier is completely dissolved in the system. The mixing process is carried out until the mass reaches room temperature. In this case, as a result of emulsification, an emulsion concentrate of the water/oil type is obtained. The resulting emulsion has a yellowish-gray color and has an odor characteristic of sea buckthorn fruits. The claimed method is demonstrated in the following example. Obtaining an emulsion concentrate of the water/oil type containing both hydrophilic and lipophilic properties of biologically active substances of sea buckthorn fruits, 1.5 kg of sea buckthorn fruits after cleaning and washing from impurities is carried through a juicer and juice is obtained. The resulting juice is filtered. The volume of the obtained juice is 1 liter. To prepare emulsions from juice, 400 ml of juice is placed in a 2-liter beaker, 800 ml of refined olive oil is added to it and, stirring constantly, the temperature of the mass is brought to 80 °C. After that, 133 g of T-2 emulsifier is added to this mass and stirred with a mixer until the emulsifier is completely dissolved in the system. The mixing process is carried out until the mass reaches room temperature. In this case, as a result of emulsification, an	emulsion concentrate of the water/oil type is obtained. The resulting emulsion has a yellowish-gray color and has an odor characteristic of sea buckthorn fruits. Technical/Commercial Advantages or Challenges As a result, the viscosity of the system is reduced, thereby increasing the solubility of biologically active substances, as well as achieve good mixing of the components and create a stable system. References 01. Flora of Azerbaijan, Baku: publishing house of the Academy of Sciences of the Azerbaijan SSR, 1955, volume VI, p. 307-309 02. Method for producing emulsion cosmetic concentrate of water/oil type from sea buckthorn fruits, EA 039564 B1.
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Source: YT SHIM (2024)

(C) Two Target Patents Selected by Two Teams of Trainees

The selected six technologies will be autonomously chosen by the teams after the selection of participants, and basic patent information and technical content will be shared.

After six Azerbaijan trainees were selected and two teams were set up (each team with three trainees), the above 6 projects were distributed. Team 1 finally selected Project 6 (A method of manufacturing concentrated emulsion for cosmetics from Azerbaijan Buckthorn fruits), while Team 2 selected the Project 1 (Eco-friendly organic fertilizers made from municipal garbages and oil/gas waste water).

1.1.2 Selection of Trainee(Pre-Instructors) for the Azerbaijan I-Corps TTT Program

In selecting instructors for TTT, it is necessary to diversify the pool to include university professors, managers of technology startup support institutions, or industry experts, similar to Korea.

The training program will be composed of both online and offline methods, and Azerbaijani faculty will conduct team projects centered on virtual business items or suitable intellectual property items filed in Azerbaijan for the training process.

The initial selection of participants will be guided to have a structure as mentioned above, ensuring an appropriate ratio of male and female instructors, and providing guidelines to prevent personnel from the same institution from being included in the same TTT project team.

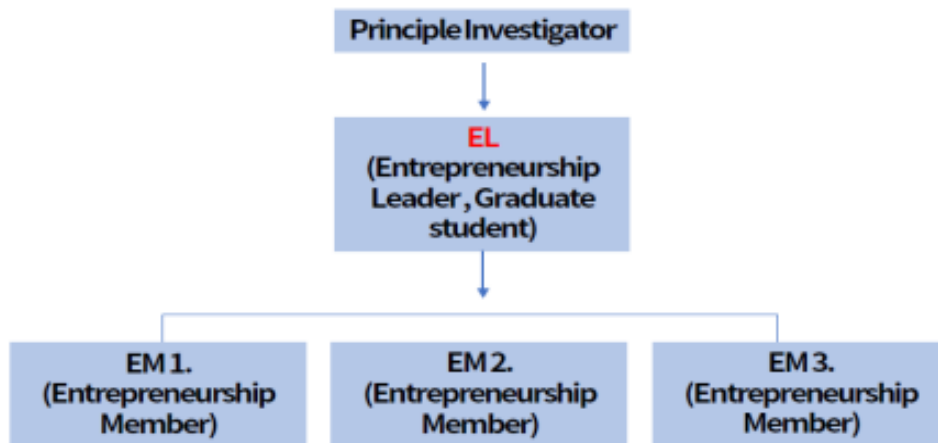
Future selection of instructors will be decided based on local circumstances in Azerbaijan, but it is recommended to strategically include personnel from startup support institutions.

The Azerbaijani faculty participating in the pilot TTT training program will consist of one EL (Entrepreneurship Leader) and two EMs (Entrepreneurship Members). However, the actual Azerbaijani I-Corps program for students will follow the same guidelines as Korea, including one EL, one EM, one supervising professor, and one student support institution representative.

However, in the case of the TTT training program, it may not be easy to operate the I-Corps program with inventors of patents and intellectual property rights participating, as in the US and Korea.

Therefore, it is recommended to design an I-Corps Team Structure similar to Korea, focusing on institutions directly conducting R&D, as the number of patents increases in the future.

[Figure 4-10] I-Corps Team Structure



Source: GS SHIM (2023)

To operate the TTT training program, four Korean instructors will be deployed in 2024. Among the four Korean instructors, one will be assigned the role of Lead Instructor, who will oversee the entire TTT training program. The remaining three instructors will provide mentoring and feedback to teams through topic-specific lectures and office hours related to the I-Corps program.

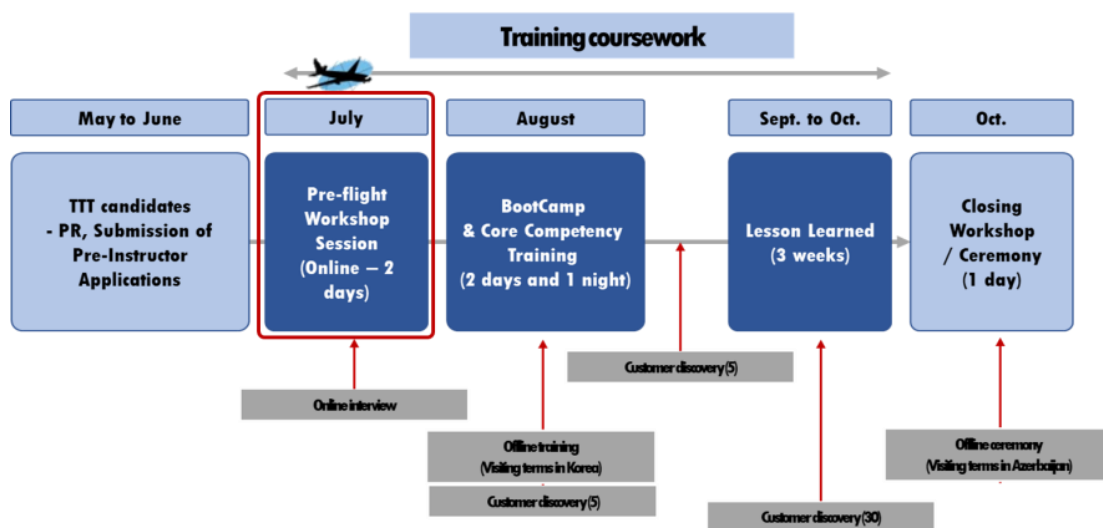
The TTT training program consists of the following four main parts:

- (Online Training: Pre-Flight) This includes lecture-based education (including orientation courses) on the core methodology of I-Corps training, the Business Model Canvas. The training aims to use practical templates and is conducted in a way that ensures real-time interaction and bidirectional communication through tools like Zoom Chat
- (Offline Training: Kick-off (BootCamp) Training) This covers understanding the I-Corps curriculum, establishing business hypotheses, Value Proposition Canvas, Customer Discovery, BM Canvas, and Beyond the Canvas. To aid in lecture delivery and curriculum understanding, supplementary educational content such as Steve Blank's free YouTube lectures can be utilized: Steve Blank: YouTube Lecture Series
- (Online Training: Office Hours) Conducted bi-weekly over six weeks, with three sessions online. Teams will perform in-depth interviews with actual customers based on the

Customer Discovery methodology learned in lectures. Each team will consist of three members (two interviewers and one recorder)

- **Office Hours:** Held bi-weekly, where instructors provide feedback on the target customers' responses gathered during Customer Discovery. Office hours are conducted using a specific PPT template developed for the I-Corps training program, rather than allowing learners to present their findings arbitrarily
- **Debrief:** Conducted immediately after office hours, where instructors discuss the Customer Discovery and team improvements without the participation of the I-Corps teams

[Figure 4-11] Training Course Design



Source: Research Team

The Korean I-Corps training program annually selects 120 to 137 teams in advance and conducts a BootCamp as group training for the selected teams. After the BootCamp, there is a one-week break, during which teams individually watch online video lectures created for pre-training purposes. The subsequent basic training course includes lecture-based education and approximately 20 Customer Discovery and Office Hours sessions to provide a comprehensive understanding of the I-Corps training program. After the basic training course, teams are given an additional 1-2 weeks to conduct approximately 25 more

Customer Discovery sessions. The advanced training course requires about 55 Customer Discovery sessions over two weeks, with weekly Office Hours. In the final week, a final presentation evaluation is conducted, and based on the evaluation results, prototype development costs are differentially supported (ranging from a minimum of 20 million KRW to a maximum of 40 million KRW per team).

For the Azerbaijan I-Corps training, it is recommended that the pre-entrepreneur program conducts approximately 100 in-depth interviews (Customer Discovery) to validate business hypotheses in the market:

- Basic training: 20 Customer Discovery sessions
- Gap period between basic and practical training: 25 Customer Discovery sessions
- Practical training: 55 Customer Discovery sessions
- Total: 100 Customer Discovery sessions

[Figure 4-12] Customer Discovery(100 times)

• **Why teams have to do 100 times of customer discovery interviews**



Source: GS SHIM (2023)

However, considering the nature of instructor training, the TTT training program will conduct approximately 20-30 interviews per team instead of 100:

- Online Education: Conducted as group training online, initially focusing on understanding the Business Model through real-time lectures

- Customer Discovery: Conducted with a goal of approximately 10 sessions per round, over six weeks with three bi-weekly sessions
- Offline Final Evaluation (in Azerbaijan): Conducted in the final week, including the results of Customer Discovery. The evaluation criteria used in the Korean I-Corps program will be provided as a reference
- Total: 30-50 Customer Discovery sessions. It is recommended that the TTT training program also conducts at least 30 Customer Discovery sessions to start deriving insights from the market

The weekly Customer Discovery sessions in the TTT training program will also receive feedback on business hypothesis validation and Customer Discovery execution through “Office Hours” with Korean instructors. Additionally, Office Hours will be conducted using the same PPT template used in the actual I-Corps training program.

The TTT training program is designed to be conducted online over a total of 4-5 weeks in 2024, with the final presentation evaluation conducted on-site in Azerbaijan in the last week, and a TTT Certificate issued upon completion of the training program.

[Figure 4-13] PPT template for Office Hours

인터뷰 결과 요약 (Summary of Customer discovery, 1st week)

■ Team name: (ABC team)

아이템 (Item)				
비즈니스 설계 (Business Thesis)	Who			
	What			
	Why			
인터뷰 (Interviews)	Total No. of Interviews	No. of interview for this week	Offline interview for this week	Online interview for this week

■ 검증한 비즈니스 가설 (Validation of your business hypothesis):

	유형 (Archetype)	가설 (Hypothesis)	검증여부
①	☒ End-user ☒ Economic payer ☒ Decision maker	☒ CPF Customer-Problem Fit OOO는 ***를 위해 @@@에 대한(로 인한) 어려움/불편함/니즈 이(가) 있음 것이다.	☒ Adopt ☒ Reject ☒ On Hold
		☒ PSF Problem-Solution Fit OOO는 @@@(불편함/문제/니즈)를 \$\$\$로 해결하고 있음(하고 싶음) 것이다.	
②	☒ End-user ☒ Economic payer ☒ Decision maker	☒ CPF Customer-Problem Fit	
		☒ PSF Problem-Solution Fit	
③	☒ End-user ☒ Economic payer ☒ Decision maker	☒ CPF Customer-Problem Fit	
		☒ PSF Problem-Solution Fit	

Source: KoEF(2023), GS SHIM (2024)

1.2 Key Guidelines for the Azerbaijan I-Corps TTT

1.2.1 2024 Azerbaijan I-Corps Training Program Plan and Key Guidelines

1.2.1.1 I-Corps TTT Training Program for Azerbaijan Educators

This training program is part of the STEPI Technology Entrepreneurship Instructor Training Program (Korean I-Corps Program, Lean Startup, and Customer Discovery Program), sponsored by the Korean government and the High Technology Park (HTP).

Although it is not an official NSF Innovation Corps program, the curriculum is based on it, and participants will complete all courses after pre-orientation and preliminary customer discovery interviews in Azerbaijan and Korea.

Through this program, the project team expects Azerbaijani educators to successfully meet the program requirements and thoroughly understand the components of the Business Model Canvas and how to conduct customer discovery interviews.

All training courses will be conducted in English, assuming that the participants are proficient in English. The STEPI educator group will perform the following roles to promote this training program:

- (Instructor) Experienced Lean Startup instructors will provide feedback on the hypotheses and assumptions included in your Business Model Canvas, interview strategies, and insights gained from lessons learned in each class. They will also provide feedback on customer discovery interviews using the interview forms provided in the course folder. For this, STEPI consists of highly experienced entrepreneurs or experts involved in fostering the venture ecosystem, who have received extensive training through the NSF and Korean I-Corps Node or Site programs
- (Program Administrator) The program administrator primarily acts as a program coordinator, providing guidance and communication throughout the program. If necessary, the program administrator will report to the program director on what needs to be done or what is required based on the participants' requests

- (Program Director) The program director will propose solutions to unforeseen problems and make necessary decisions to enhance the effectiveness of the training program in consultation with the classroom instructors

1.2.1.2 Training Program Culture

Instructors push, challenge, and question teams to learn quickly due to time constraints.

Instructors are direct, open, and rigorous in conducting the program, similar to the real world, but the main purpose of the training is to help understand the Lean Startup process.

Therefore, comments from instructors and mentors should not be taken as criticism of individuals or teams but as part of the process of learning about the process, market, and oneself.

Instructors also expect teams to ask questions, challenge their perspectives if they disagree, and engage in real conversations with the education team. This approach may sometimes feel harsh or abrupt, but it is intended to help teams learn to challenge themselves quickly and objectively and understand that they need to learn and evolve incredibly fast as entrepreneurs.

The education team strives to start and end on time, so punctuality is strictly enforced.

Therefore, all participating teams should carefully check the time allocated for presentations and may face disadvantages for being late or unprepared.

Participants must enter interview notes, revise the Business Model Canvas, and communicate with instructors by the scheduled time using online task support tools and templates such as Box.

Attendance at all sessions is mandatory, and the only exception for not attending a session is a critical customer discovery interview that cannot be scheduled at another time, except for serious illness. Absences for other reasons must be discussed with the instructors and mentors and approved in advance by the program director or instructors.

1.3 Training Program Support Tools

1.3.1 Utilization of Cloud-Based File Sharing System

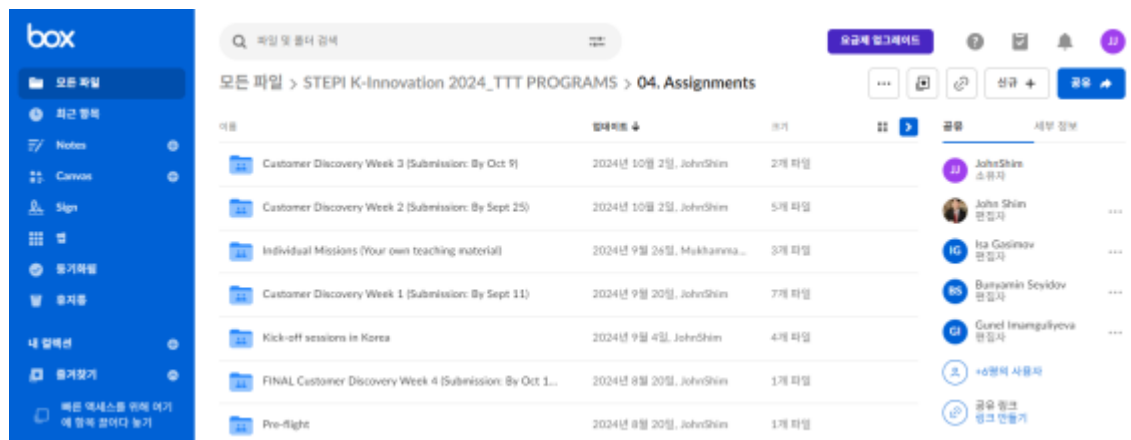
The core of this course is the interviews conducted by participating teams with potential customers and analyzing this data.

Teams should conduct customer discovery interviews outside the school or affiliated institution's buildings and spend significant time testing hypotheses about potential customers' needs, wants, and desires.

Class time focuses on key learnings obtained from conversations with customers, excluding what was already known before entering the course.

Teams should aim for at least 30 interviews by the end of the training program and are encouraged to schedule potential customer discovery interviews before the Bootcamp (Kick-off) program starts.

[Figure 4-14] Collaboration Tool(Box.com)



Source: Research Team

STEPI uses a collaboration application called “Box,” through which assignments/presentations are stored, and all course materials (e.g., schedules, lectures, presentation templates, assignment templates, etc.) are accessed.

Additionally, information on this file-sharing platform will be provided with an invitation link from STEPI Box before the Bootcamp or office hour sessions start.

Once you join and accept the invitation link for access to the file-sharing platform, you can continue to access and log in to Box through this link. (<https://www.box.com/home>)

Each assignment must be uploaded as a PDF to the appropriate session upload folder with the correct date and title, and for office hours, it must be uploaded as a PDF in the format of team number_team name_session number according to the instructions provided via email.

For files that need to be submitted in advance for office hours, detailed instructions on the assignment and necessary assignment templates will be provided the week before, and all assignments must be submitted by midnight (11:59 PM AZT) on the Wednesday of the week of the next session presentation.

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CHAPTER 5

Conclusion





Chapter 5. Conclusion

The implementation of the application process, along with capacity building within the TT and TC systems in Azerbaijan, has been a significant milestone in fostering entrepreneurship and innovation in the country. This project has not only addressed the critical gaps in the ecosystem but has also paved the way for a sustainable and robust infrastructure to support emerging businesses. By integrating Korean best practices with local insights, the initiative has successfully created a foundation for fostering innovation, increasing economic diversification, and promoting the development of a knowledge-based economy.

One of the project's key achievements has been the establishment of a streamlined, transparent, and accessible application process. By implementing digital tools and platforms, the application process was made more efficient, user-friendly, and inclusive. This digital transformation has enabled aspiring entrepreneurs to submit their proposals and access incubation services regardless of their geographical location. Additionally, the standardized evaluation criteria and merit-based selection processes ensured that the most promising ideas and ventures were given the opportunity to flourish.

The application process's refinement has also contributed to reducing barriers to entry for underrepresented groups, including women, youth, and individuals from rural areas. This inclusivity has been instrumental in fostering a diverse pool of innovators, enriching the ecosystem with varied perspectives and ideas. Furthermore, the integration of feedback mechanisms allowed the system to evolve continuously, addressing user concerns and adapting to emerging trends.

The project's emphasis on capacity building has been pivotal in enhancing the skills and competencies of stakeholders within the start-up and TC ecosystem. Training programs, workshops, and mentorship initiatives were organized to equip entrepreneurs, incubator managers, and policymakers with the knowledge and tools required to succeed in their respective

roles. These initiatives bridged critical skill gaps, enabling stakeholders to navigate challenges more effectively and leverage opportunities in the fast-paced world of innovation and technology.

Moreover, partnerships with Korean organizations and experts have facilitated knowledge transfer and the adoption of global best practices. By fostering collaborations between local and international stakeholders, the project has enriched the ecosystem with diverse expertise and perspectives. These partnerships have also strengthened Azerbaijan's position in the global innovation landscape, creating opportunities for cross-border investments, market access, and collaborative ventures.

The outcomes of this initiative are already evident in the form of increased TC activity, improved incubation services, and heightened interest in entrepreneurship across Azerbaijan. By nurturing innovative ventures, the project has contributed to job creation, economic diversification, and the development of cutting-edge solutions to societal challenges.

Looking forward, the sustainability of these achievements depends on continuous investment in the start-up system. Policymakers must ensure that the regulatory environment remains conducive to innovation, while private sector involvement should be encouraged to diversify funding sources and foster a culture of entrepreneurship. Additionally, efforts to promote entrepreneurship education and awareness at all levels will further strengthen the pipeline of innovative talent.

In conclusion, the successful implementation of the application process and capacity-building initiatives marks a significant step toward establishing Azerbaijan as a regional hub for innovation as well as TC. The project's holistic approach, combining technology, inclusivity, and international collaboration, has set a strong precedent for future initiatives. By sustaining and building upon these efforts, Azerbaijan can continue to unlock the potential of its TC ecosystem, driving long-term economic growth and societal progress.

The project can also be considered a new stage in the scientific, technological, and innovation relations between Azerbaijan and Korea. We hope that this project, implemented during 2022-2024, will pave the way for the establishment of mutual relations, the emergence of new initiatives, and the implementation of new projects as a result.

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